UniSpark: Exploring Hybrid Attention for Multimodal Understanding and Generation

Jiancheng Pan*, **Yuzheng Liu***, **Hengyuan Cao***, **Qiyuan Zhang***, **Yangyan Li**, **Biao Gong**†

*Equal Contribution †Project Lead

Abstract

We present **UniSpark-9B**, a unified multimodal model that explores hybrid attention for understanding and generation, with approximately 9B parameters in each of its understanding and generation branches. UniSpark combines full attention and GatedDeltaNet linear attention within a shared Mixture-of-Transformer-Experts backbone. Full-attention layers capture explicit token interactions, while recurrent layers process sequences with linear attention cost and maintain fixed-size context states. This combination provides a shared basis for efficient multimodal understanding and generation. Autoregressive text prediction and visual flow matching support visual question answering, image generation, instruction-guided editing, and interleaved prediction. To organize multimodal context, the model separates private noisy visual targets from reusable clean content through coordinated attention masks and recurrent state updates. Our training-data sources include approximately 159.7M generation samples, 213.4M understanding samples, and 624.8K interleaved sequences, together with image-editing and text-only data. Understanding data cover captioning, visual question answering, OCR, grounding, and GUI tasks. Event-driven video annotation pairs observed keyframes with scene descriptions and transition instructions to construct interleaved sequences. Task-aware packing preserves sample boundaries and multimodal structure throughout a training recipe that moves from image resolution scaling and supervised refinement to video generation and editing through post-SFT adaptation. By combining image and video tasks during adaptation, the model extends its visual generation capabilities to motion and video editing within the same hybrid-attention framework.

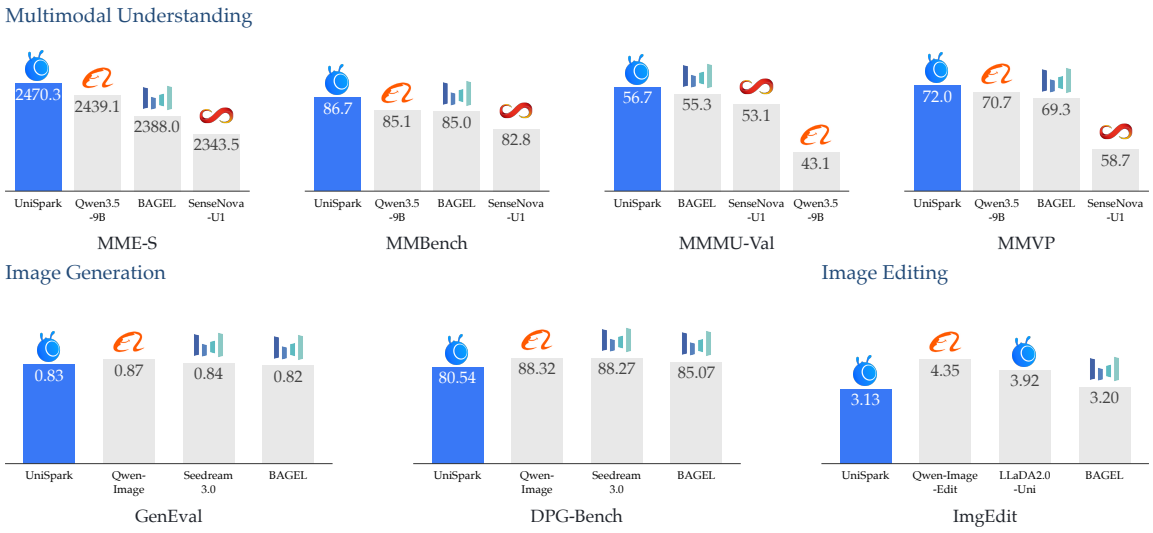


Figure 1: **Multimodal understanding, generation, and editing performance.**

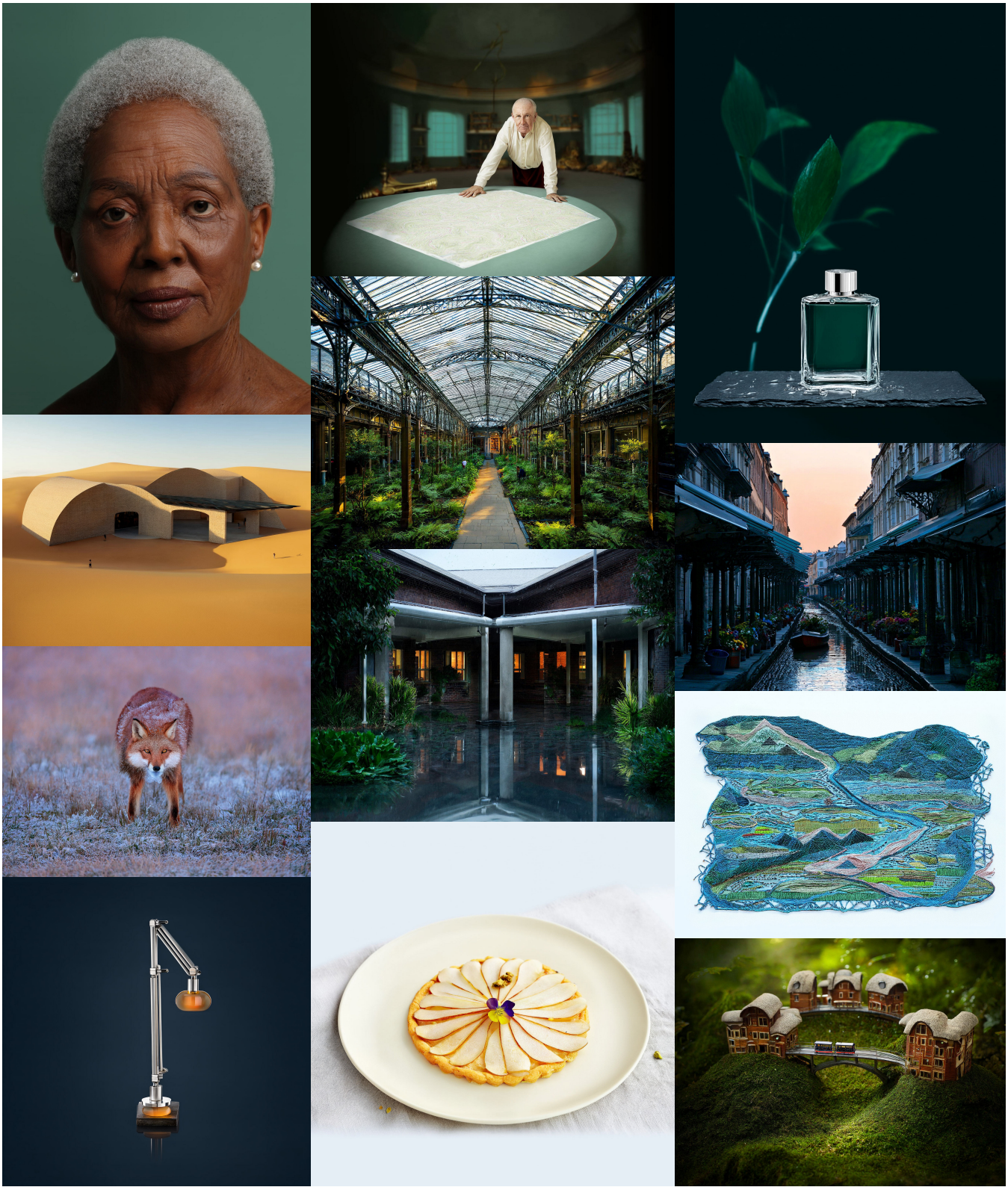
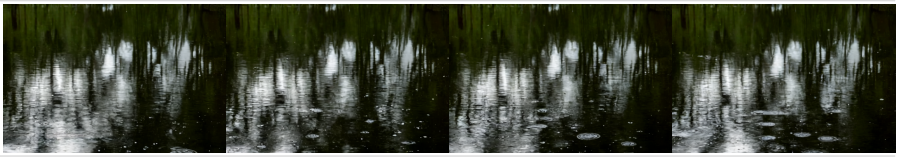


Figure 2: **Text-to-image generation.**

Ocean waves wash against dark coastal rocks beneath a pale overcast sky. Fixed camera; preserve the rocky shoreline and horizon.
Waves and white sea foam continue moving around the same rocks.

An orange tabby kitten plays with a blue yarn ball on a beige rug. Fixed camera and consistent subjects.
Lift a paw, tap the ball and continue playing.

A quiet pond reflects green trees under soft overcast light. Fixed camera and the same pond surface.
Raindrops strike the water and form overlapping circular ripples.



A fixed-camera close photograph of a white ceramic cup filled with hot tea on a dark wooden table. Soft side light, plain dark background; preserve the cup and table.
Wisps of steam curl upward from the hot tea. Continue the steam rising above the same cup.

A green parrot with a red forehead perches on a wooden branch against a warm gray background. Fixed camera, soft light, same bird and perch.
Raise the folded wings, stretch them outward, then fold them back.

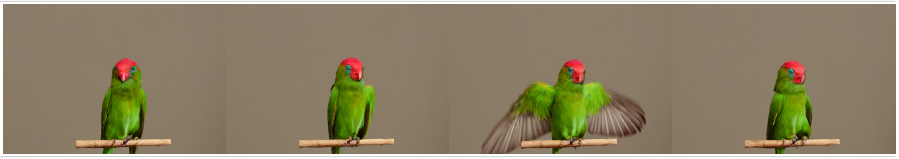


Figure 3: **Image editing and interleaved generation.**

Contents

1 Introduction **5**

2 Related Work **6**

2.1 Unified Multimodal Understanding and Generation 6

2.2 Efficient and Hybrid Attention Mechanisms 7

3 Model Design **7**

3.1 Unified Multimodal Representations 7

3.2 Hybrid-Attention Mixture-of-Transformers 9

3.3 Joint Understanding and Generation Objectives 11

3.4 Interleaved Generation with Persistent Context 11

4 Data Preparation **12**

4.1 Multimodal Understanding 12

4.2 Image Generation 14

4.3 Image Editing 15

4.4 Interleaved Data 15

4.5 Text-Only Data 16

4.6 Video Data 16

4.7 Multimodal Sequence Packing 17

5 Training Strategy **18**

5.1 Joint Multimodal Foundation 19

5.2 Progressive Resolution Scaling 19

5.3 High-Resolution Refinement 19

5.4 Supervised Capability Refinement 19

5.5 Optimization 20

5.6 Adaptation to Video Generation and Editing 20

5.7 Training Infrastructure and Efficiency 20

6 Experiments **21**

6.1 Evaluation Setup and Baselines 21

6.2 Multimodal Understanding 21

6.3 Text-to-Image Generation 22

6.4 Instruction-Based Image Editing 23

6.5 Interleaved Generation and Sequence Editing 23

6.6 Extension to Video Generation and Editing 23

6.7 Inference Efficiency and Scaling 24

7 Conclusion **28**

1 Introduction

Multimodal understanding and visual generation place complementary demands on a model. Understanding requires extracting evidence from images and expressing it in language. Generation requires turning language and visual conditions into spatially coherent content, while editing must also preserve the relevant parts of a reference image. Combining autoregressive language modeling with continuous visual prediction has become a central approach to unification, as demonstrated by Transfusion and Show-o2 (Zhou et al., 2024; Xie et al., 2025). BAGEL brings understanding, generation, and editing into a shared multimodal sequence and backbone through modality-specific transformer parameters (Deng et al., 2025), while SenseNova-U1 further explores native unified representations and interleaved vision-language generation (Diao et al., 2026). These developments motivate studying not only individual image outputs, but also sequences in which completed outputs become context for subsequent predictions.

Hybrid attention introduces a further design question. Full-attention layers retain explicit interactions with preceding tokens (Vaswani et al., 2017), whereas recurrent linear-attention layers summarize context in a persistent state (Katharopoulos et al., 2020; Yang et al., 2025b). Kimi Linear and Olmo Hybrid investigate layer-wise combinations of recurrence and attention, providing evidence for the potential of hybrid architectures in language modeling (Kimi Team et al., 2025b; Merrill et al., 2026). Qwen3.5 provides a backbone combining full attention with GatedDeltaNet layers (Qwen Team, 2026). Our motivation is to examine whether a hybrid-attention backbone can jointly support multimodal understanding and visual generation, despite their different prediction and context-update requirements. These operators must accommodate both causal text prediction and iterative visual denoising, while higher resolutions and multiple images increase the visual token workload. We investigate this question within a unified framework and characterize its capabilities and computational trade-offs, providing empirical evidence to inform future hybrid multimodal model design.

Interleaved generation makes the context problem particularly concrete. UniAR identifies shared visual context as a bridge between interpreting and producing images, enabling direct reuse of generated discrete visual tokens (Peng et al., 2026). For continuous visual generation, the corresponding challenge is deciding when a prediction is ready to become context. Consider a model that produces an image, explains it, and then edits it in response to a subsequent instruction. The first image is initially a noisy prediction target, but its completed content must later become a reusable condition. If intermediate denoising updates enter persistent context, later predictions can depend on temporary noise rather than completed visual content. Conversely, discarding all visual history would remove information needed to maintain references across outputs. A unified sequence representation must distinguish these roles, and each attention operator must implement the distinction according to its own update rules.

We present **UniSpark**: Exploring Hybrid Attention for Multimodal Understanding and Generation. Our UniSpark-9B configuration uses approximately 9B parameters in each of the understanding and generation branches; the 9B label refers to the size of each branch. UniSpark builds on the Mixture-of-Transformer-Experts formulation (Liang et al., 2025; Deng et al., 2025) and the Qwen3.5 hybrid-attention backbone (Qwen Team, 2026). Text and semantic visual features use understanding parameters, while VAE latents use generation parameters. These streams occupy one packed sequence and exchange context through the shared backbone. Autoregressive text prediction and visual flow matching (Lipman et al., 2023) jointly support multimodal understanding, text-to-image generation, reference-conditioned image editing, and interleaved prediction within a shared architecture, without requiring a separate task-specific generator.

The information-flow design separates *private prediction targets* from *public context*. A noisy visual block reads its public prefix but is hidden from later blocks. Full-attention layers implement this rule with block visibility masks and allow interactions within a visual target. Linear-attention layers process the block recurrently, but discard its private state and convolution-history updates after local computation on a noisy block. Clean visual representations may instead update public context. In interleaved training, the clean representations come from observed images through teacher forcing (Williams & Zipser, 1989); in a generated sequence, they must come from completed model outputs. This creates the familiar distinction between conditioning on observed history during training and model-generated history during inference (Bengio et al., 2015).

The data and training design exposes this backbone to several forms of supervision. Understanding data cover captioning, visual question answering, OCR, GUI tasks, and grounding. Generation, editing, interleaved sequences, and text-only examples use the same sequence-planning interface. Online packing combines examples under token budgets while preserving sample boundaries, attention blocks, and loss locations.

The selected sources contain 159.7M generation samples and 213.4M understanding samples, including 10.0M

OCR and 4.5M grounding samples, as well as 624.8K interleaved sequences. A separate video-annotation source list provides 24.2M source samples for constructing additional event-driven sequences. This pool is excluded from the reported interleaved training count. The data extend beyond image–caption pairs to support general semantic learning, specialized visual tasks, and the modeling of state changes over time.

Our training recipe develops image capabilities in four stages before adapting the model to video generation and editing. Joint Multimodal Foundation establishes the shared prediction interface at low resolution. Progressive Resolution Scaling and High-Resolution Refinement increase spatial detail up to 2K, while Supervised Capability Refinement improves understanding, instruction following, and image generation and editing. These four stages share the joint text and visual objectives. The subsequent video adaptation mixes image and video tasks while keeping the understanding branch frozen, carrying the learned image capabilities into video generation and editing. Hybrid sharding, activation checkpointing, and variable-length attention execution support the changing workload as visual sequences grow.

Our main contributions are:

- **Hybrid multimodal modeling.** A shared MoT backbone combines full attention and recurrent linear attention with separate text and visual prediction objectives, connecting understanding and generation through common multimodal context.
- **Context handling for interleaved prediction.** Explicit public/private block roles separate noisy visual targets from reusable clean content and specify how visibility, recurrent state, and context insertion interact across successive outputs.
- **Task-aware data and progressive training.** A common packing interface organizes 159.7M listed generation samples, 213.4M understanding samples, and 624.8K interleaved sequences. Resolution scaling and supervised refinement build image capabilities up to 2K, which are extended to video generation and editing through post-SFT adaptation with mixed image and video data.
- **Efficient hybrid multimodal architecture.** Recurrent linear-attention layers provide linear attention cost in sequence length and fixed-size context states, reducing attention computation and context-storage requirements in these layers for both understanding and generation.

2 Related Work

2.1 Unified Multimodal Understanding and Generation

Unified multimodal models use a shared model to understand and generate visual content. Transfusion combines autoregressive text prediction with image diffusion (Zhou et al., 2024). Mixture-of-Transformers uses separate parameters for each modality while sharing attention across them (Liang et al., 2025). BAGEL builds on this approach and trains a unified model on interleaved multimodal data (Deng et al., 2025). These works form the basis of our shared architecture and joint text and visual training objectives.

Other models explore how to represent images and support more tasks. Tuna-2 uses direct pixel-patch embeddings in place of separate visual encoders (Liu et al., 2026). SenseNova-U1 supports understanding, image generation, and interleaved vision-language generation with dense and MoE backbones (Diao et al., 2026). SenseNova-Vision handles different vision tasks through text, image, or mixed outputs without separate prediction heads (Han et al., 2026). UniSpark uses semantic visual features and VAE latents, and studies how hybrid attention lets these representations share context.

Sharing visual context also allows one output to guide the next. UniAR uses a shared discrete visual tokenizer so that generated visual tokens can be read without encoding the image again (Peng et al., 2026). Context Unrolling studies reasoning through text, images, video, and geometry before making predictions (Yang et al., 2026). For continuous visual generation, UniSpark keeps temporary noisy predictions separate from reusable clean content. Attention masks and recurrent-state rules control which information later outputs can use. Training uses observed clean images as context, while inference uses completed model outputs.

A model that supports both tasks does not necessarily improve one task by learning the other. VGAU-Diag tests when generated visual aids help understanding and finds cases where models fail to use even correct visual aids (Zhu et al., 2026). Wu et al. examine how understanding and generation help each other at the representation, task, and system levels, and identify task specialization and shared knowledge as important factors (Wu et al., 2026). These studies show why unified models need to be evaluated on both their individual tasks and their use of shared visual information.

2.2 Efficient and Hybrid Attention Mechanisms

Recurrent attention stores past information in a state that is updated as tokens are processed. Gated DeltaNet uses adaptive forgetting and delta-rule updates (Yang et al., 2025b). Kimi Linear adds finer gating through Kimi Delta Attention and combines it with Multi-Head Latent Attention across layers (Kimi Team et al., 2025b). Gated DeltaNet-2 uses separate erase and write gates for each memory channel (Hatamizadeh et al., 2026). These methods control how memory is updated. UniSpark adds a block-level rule that decides whether updates from a noisy visual target can enter the context used by later predictions.

Hybrid backbones combine recurrent layers with full attention. UniSpark starts from Qwen3.5, which combines GatedDeltaNet and full-attention layers (Qwen Team, 2026). Olmo Hybrid studies this combination through theory and controlled language-modeling experiments (Merrill et al., 2026). In vision, SoLA-Vision studies how to arrange linear and softmax attention across layers (Li et al., 2026). SANA-Video 2.0 combines gated linear attention, periodic softmax layers, and attention residuals for video diffusion (Chen et al., 2026). Our work explores hybrid attention in a model trained jointly for multimodal understanding and generation.

Attention types can also be combined within a layer. HydraHead assigns full and linear attention to different heads, keeps full attention for heads important to retrieval, and normalizes their outputs before combining them (Tan et al., 2026). UniSpark combines the two attention types across layers. Reusing context also requires careful state handling: HARTS studies shared prefixes in branching agent rollouts, including state transfers and activation recomputation (Meng et al., 2026). Although HARTS focuses on agent reinforcement learning, it illustrates the need to coordinate recurrent states when computation branches or context is reused.

UniSpark brings these ideas to a shared multimodal backbone with separate understanding and generation experts. Attention masks and recurrent-state rules keep noisy prediction blocks separate, while clean visual content remains available to guide later outputs.

3 Model Design

UniSpark explores hybrid attention for multimodal understanding and generation. The central question is how a backbone combining recurrent linear attention and full attention can support both causal language prediction and iterative visual generation, particularly when text and images are interleaved. These tasks impose different requirements on information flow: text predicts the next token from its prefix, visual generation refines a block of noisy latent tokens, and subsequent outputs must condition on the completed visual content without inheriting intermediate noise states.

3.1 Unified Multimodal Representations

We build on the multimodal Mixture-of-Transformer-Experts (MoT) formulation of BAGEL (Deng et al., 2025) and the hybrid attention backbone of Qwen3.5 (Qwen Team, 2026). Text, semantic visual features, and visual latents occupy a common sequence, with separate understanding and generation parameters inside each layer. We study how these experts exchange context through two different attention operators, and how their context persists across visual generation boundaries. Fig. 4 summarizes this organization. The model combines unified representations, a hybrid MoT backbone, joint prediction objectives, and an explicit protocol for interleaved visual context.

Model scale. UniSpark-9B uses approximately 9B parameters in each of its understanding and generation branches. The 9B label refers to the backbone size of each branch, rather than the total parameter count of the complete unified system.

A multimodal example is represented as an ordered sequence of text and visual blocks, together with conversation roles, media boundaries, and supervision indicators. Text is mapped through the language embedding layer. Images and video frames used for understanding are encoded by the Qwen3.5 vision encoder, whose output features enter the language hidden space. Visual content used for generation is encoded by a VAE, patchified, and mapped through a latent adapter. Semantic features provide an understanding-oriented representation, while VAE latents retain the representation used by the visual prediction objective. Predicting in a learned image-latent space follows the latent generative modeling approach of Rombach et al. (Rombach et al., 2022).

Visual inputs and targets. Let $\mathbf{z}_0$ denote a patchified clean visual latent and $\mathbf{z}_t$ its noisy counterpart. The generation-side input takes the form

$$\mathbf{h}_t^{\text{vis}} = A_{\text{in}}(\mathbf{z}_t) + E_{\text{time}}(t) + E_{\text{pos}}(\mathbf{p}), \tag{1}$$

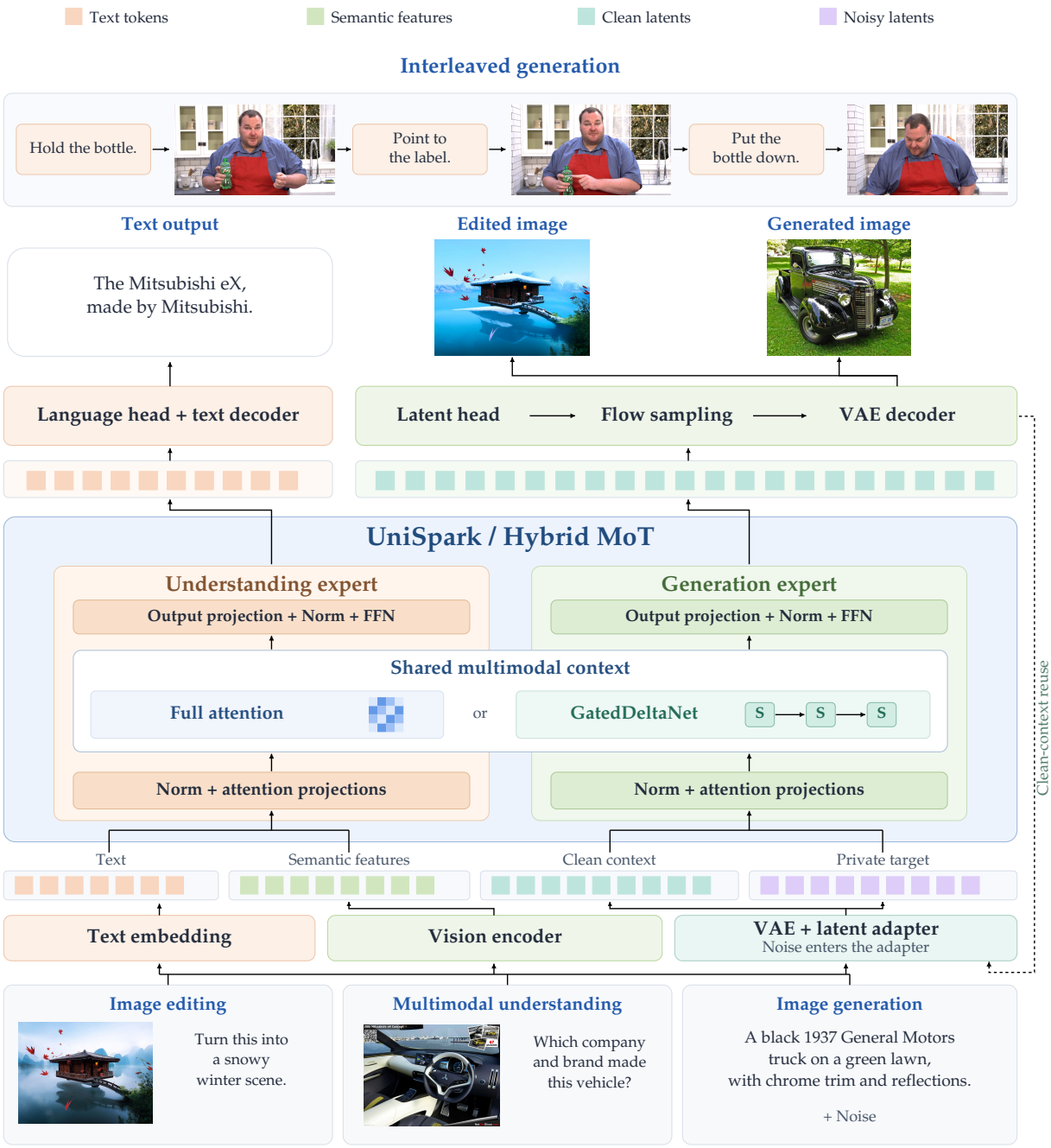


Figure 4: **Architecture of UniSpark.** Understanding and generation experts share multimodal context through full and linear attention, supporting text prediction, image generation, editing, and interleaved outputs.

where A_{in} projects latent patches to the backbone width, and $\mathbf{p}$ encodes their spatial or spatiotemporal locations through an additive latent position embedding. The media layouts preserve these coordinates, while multimodal rotary positions organize text and visual blocks within the full-attention layers. These two positional mechanisms thus serve distinct roles: the additive embedding encodes locations within each medium, whereas rotary positions structure the multimodal sequence. Clean visual context is assigned $t = 0$, while noisy prediction targets are assigned $t > 0$ and receive visual supervision.

Image and video configurations. The representation interface separates the visual codec from the shared backbone. The unified configuration uses the Wan2.2 VAE (Wan Team, 2025) for both images and videos, treating an image as a single-frame video and sharing the latent adapters and three-dimensional position embedding. Codec, modality, and role embeddings distinguish visual blocks in the image–video configuration. The interleaved construction below focuses on text and images.

3.2 Hybrid-Attention Mixture-of-Transformers

Specialized parameters with shared context. The backbone assigns text and semantic visual tokens to an understanding expert, and VAE latent tokens to a generation expert. Each decoder layer contains separate normalization and feed-forward parameters for the two experts, together with expert-specific attention projections. Routing follows token type. Despite these expert-specific parameters, full attention is computed jointly over all tokens in the multimodal sequence. VAE latent tokens can therefore attend to text and semantic visual tokens at each layer, allowing the generation expert to condition on both.

Complementary attention operators. The layer schedule combines full attention with GatedDeltaNet linear attention (Yang et al., 2025b). A full-attention layer computes explicit query–key interactions over the permitted context. Within a visual target block, this provides bidirectional spatial interaction. A linear-attention layer summarizes preceding information in a recurrent matrix state. For fixed head dimensions, its recurrent computation grows linearly with sequence length, although the complete hybrid model still contains full-attention layers. This combination provides a setting for examining the tradeoff between explicit visual interactions and compressed context propagation as interleaved sequences become longer.

For clarity, a single-head gated delta update can be written as

$$\begin{aligned} \overline{\mathbf{S}}_i &= \alpha_i \mathbf{S}_{i-1}, \\ \mathbf{S}_i &= \overline{\mathbf{S}}_i + \beta_i \mathbf{k}_i \big(\mathbf{v}_i - \overline{\mathbf{S}}_i^\top \mathbf{k}_i\big)^\top, \\ \mathbf{o}_i &= \mathbf{S}_i^\top \mathbf{q}_i, \end{aligned} \tag{2}$$

where i indexes tokens, α_i controls state decay, and β_i controls the delta update. The equation omits head-wise normalization, output gating, and projection details. Understanding and generation tokens use their respective parameters to produce these quantities, but update a shared state along the permitted multimodal history. The short convolution history is likewise shared across expert boundaries. Thus, a change of expert does not by itself reset the accumulated context.

Public context and private visual targets. Hybrid attention requires both a visibility mask and a state-update policy. We distinguish causal public blocks, clean visual public blocks, and private noisy target blocks. In full-attention layers, a causal block sees the earlier public prefix and its own causal prefix. A clean visual block or a noisy target block sees the earlier public prefix and its entire own block. Other blocks cannot attend to a private target, including its intermediate noisy representations. Different packed examples are isolated.

Linear-attention layers use the same block boundaries but implement them through recurrent state propagation. Let $\mathcal{R}(B, \mathbf{S})$ denote processing block B from state $\mathbf{S}$. The public state obeys

$$\mathbf{S}_{b+1}^{\text{pub}} = \begin{cases} \mathcal{R}(B_b, \mathbf{S}_b^{\text{pub}}), & B_b \text{ is public}, \\ \mathbf{S}_b^{\text{pub}}, & B_b \text{ is private}. \end{cases} \tag{3}$$

A private block reads the public prefix and performs local updates to produce its outputs, but its final recurrent state and convolution history are discarded. Public blocks retain both as public context. This prevents one denoising target from changing the context used by later predictions.

The two operators do not have identical within-block behavior. Linear attention retains causal recurrence even within a block labeled full or private, whereas full attention permits bidirectional interaction there. They agree on the public/private boundary, while full-attention layers supply the explicit within-target interactions absent from the recurrent operator.

Fig. 5 connects these boundaries to UniSpark's task formats. In multimodal understanding, a clean image and a question form public context for autoregressive answer prediction. Text-to-image generation uses a public prompt to condition a private noisy visual target. Image editing additionally places the clean source image and edit instruction in the public prefix. Interleaved prediction repeats these roles: for $[T_1, N_1, C_1, T_2, N_2]$, later text T_2 and the next visual target N_2 can use clean context C_1, but not the private target N_1. The earlier

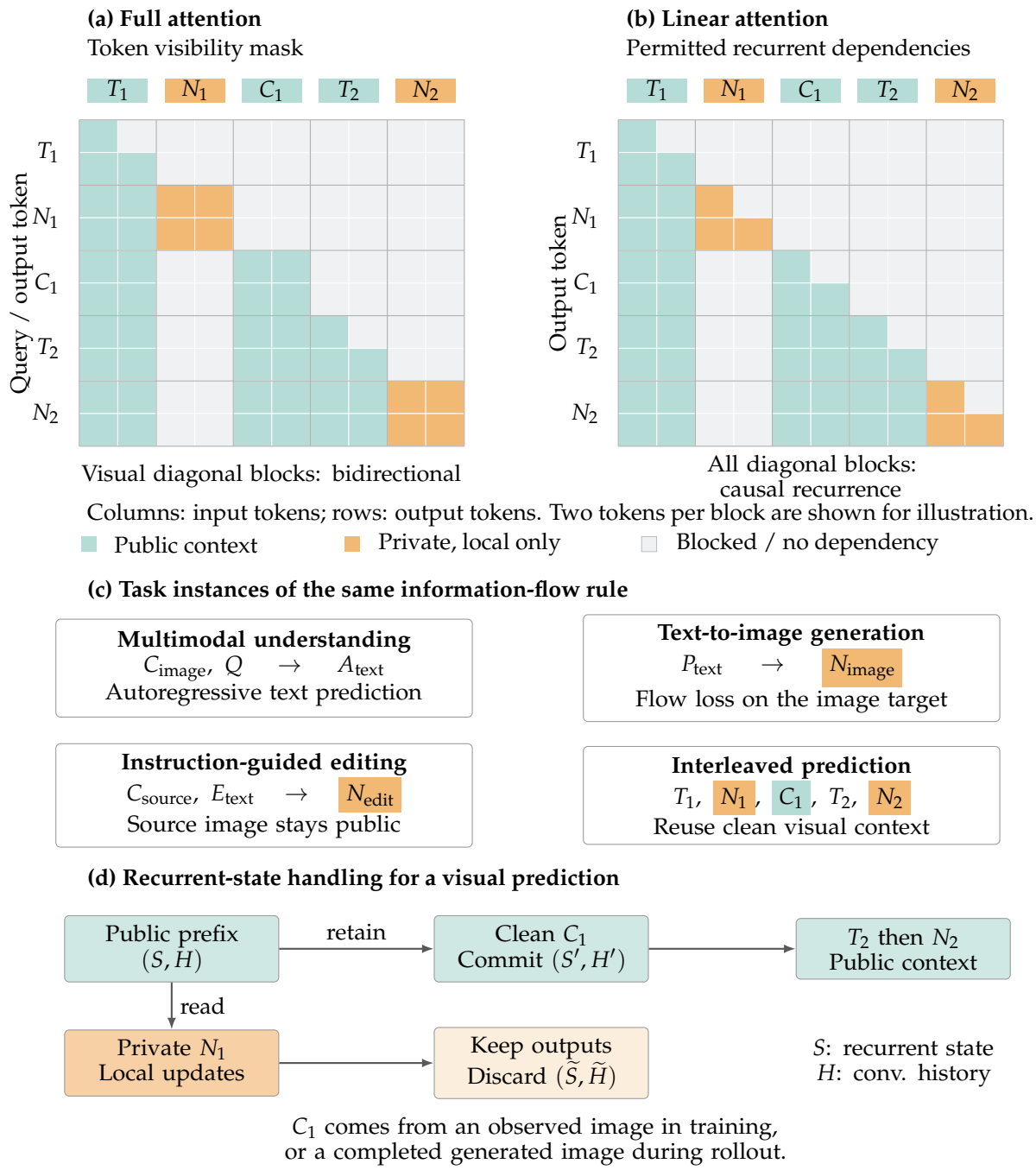


Figure 5: **Information flow under hybrid attention.** Visibility masks and recurrent-state rules preserve the same public/private boundary, allowing clean context reuse without propagating noisy targets.

target cannot read its later clean counterpart either. Text outputs receive autoregressive supervision where selected by the training plan, while noisy visual targets receive flow-matching supervision.

The matrices distinguish direct token visibility in full attention from permitted recurrent dependencies in linear attention. The latter is not an explicitly constructed attention mask. During visual prediction, local recurrent and convolution updates produce the target outputs without changing the public prefix; separately processing a clean image makes its content available to subsequent understanding or generation.

3.3 Joint Understanding and Generation Objectives

The same backbone produces language logits and visual velocity predictions. For supervised text positions $\mathcal{T}$, the language objective is

$$\mathcal{L}_{\text{text}} = -\mathbb{E}\left[\frac{1}{|\mathcal{T}|}\sum_{i\in\mathcal{T}}\log p_\theta(w_i \mid \mathcal{C}_{<i})\right], \tag{4}$$

where $\mathcal{C}_{<i}$ is the permitted multimodal prefix, and the supervision mask selects the desired assistant outputs. Visual inputs can therefore inform text prediction without making future text visible.

Visual generation uses flow matching (Lipman et al., 2023). Given noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a sampled time $t \in (0, 1)$, we construct

$$\mathbf{z}_t = (1-t)\mathbf{z}_0 + t\boldsymbol{\epsilon}, \qquad \mathbf{u}_t = \boldsymbol{\epsilon} - \mathbf{z}_0. \tag{5}$$

The output adapter maps the target-token hidden states to a velocity field, which is supervised by

$$\mathcal{L}_{\text{vis}} = \mathbb{E}_{\mathbf{z}_0,\boldsymbol{\epsilon},t}\left[\|\widehat{\mathbf{u}}_\theta(\mathbf{z}_t, t; \mathcal{C}) - (\boldsymbol{\epsilon} - \mathbf{z}_0)\|_2^2\right]. \tag{6}$$

Here $\mathcal{C}$ contains the public conditions preceding the target. Only target latents receive this loss; clean context latents do not. Text and visual terms are combined as $\mathcal{L} = \lambda_{\text{text}}\mathcal{L}_{\text{text}} + \lambda_{\text{vis}}\mathcal{L}_{\text{vis}}$. The expressions specify the prediction targets; task weighting, loss normalization across packed examples, and the timestep distribution are training choices. At inference, visual sampling integrates the predicted field from noise at $t = 1$ toward data at $t = 0$.

Conditional generation and editing. Text-to-image generation conditions a private visual target on text. Image-conditioned generation and editing additionally supply clean reference-image representations in the public prefix. Semantic features and clean VAE latents can both represent a reference, exposing its content to the shared backbone through the understanding and generation experts, respectively. The instruction, reference representations, and noisy target are organized by their roles in the same sequence. This formulation also allows an earlier image in an interleaved example to become a condition for a later visual prediction.

3.4 Interleaved Generation with Persistent Context

Interleaved generation extends conditional prediction to an ordered sequence of text spans and visual outputs. For blocks $B_1, \ldots, B_K$, the modeling factorization is

$$p_\theta(B_1, \ldots, B_K) = \prod_{k=1}^{K} p_\theta(B_k \mid B_{<k}), \tag{7}$$

where a text block is predicted autoregressively and a visual block is modeled by conditional flow matching. This factorization specifies the conditioning order, not a tractable likelihood evaluation for the visual flow objective. Later text can describe an earlier image, and later images can use preceding text and visual content as conditions. Illustrated explanations, sequences of edits, and visually grounded multi-step responses are potential applications of this sequence structure.

From a prediction target to reusable context. A visual output has two distinct roles during interleaved training. Its noisy latent is a private prediction target. Its clean representation is a public condition for subsequent outputs. A schematic two-image training sequence is

$$[T_1,\ N_1(t_1),\ C_1,\ T_2,\ N_2(t_2)], \tag{8}$$

where T_k is a text span, N_k is a private noisy visual target, and C_1 contains clean representations of the first image. The second target can read T_1, C_1, and T_2, but not N_1. Likewise, N_1 cannot read its own later clean context C_1. Clean context may contain both VAE latents and ViT features, associated with the same semantic image identity. Their positions are assigned consistently with the target's media span.

The clean context in training comes from the observed image, using teacher forcing. The context copy has no visual denoising loss, while the preceding private copy carries the flow objective. This separation lets multiple visual targets coexist in a packed example without exposing the noise in an earlier target to later outputs. During inference, clean context instead comes from the model's completed images.

Context updates during decoding. A multi-image rollout requires alternating text prediction, visual sampling, and context updates. During sampling of one image, each denoising evaluation reads the established public prefix without retaining its intermediate target state in public context. Once the image is completed, its clean visual representations can be encoded and appended as public context before the next text or image block. Full-attention layers retain the corresponding public key–value entries, while linear-attention layers retain the updated recurrent state and convolution history. Private denoising history must not be reused as the clean image's persistent context.

The inference interface accepts interleaved text–image conditions and produces one image per call, optionally preceded by generated text. Multi-image decoding requires a controller to repeat sampling and clean-context insertion and determine when to stop.

Interleaving makes the hybrid-attention question concrete: explicit attention must support visual interactions within each target, recurrent layers must carry useful multimodal context across blocks, and both must respect the boundary between temporary noise and completed content. This organization provides the architectural basis for examining understanding, conditional generation, and interleaved prediction within UniSpark.

4 Data Preparation

UniSpark combines data for multimodal understanding, image generation, image editing, text-only learning, and interleaved prediction during its main training phase. We organize these sources by their supervision structure, while adjusting resolution and sampling separately across training stages. The curriculum comprises *Joint Multimodal Foundation*, *Progressive Resolution Scaling*, *High-Resolution Refinement*, and *Supervised Capability Refinement (SFT)*. The first three stages establish joint modeling and extend visual generation to resolutions up to 2K. SFT targets more accurate understanding, better instruction following, and improved visual generation and editing. Separately, we collect video generation and editing data to further fine-tune the model, as an exploratory post-SFT effort to develop video capabilities. The stage objectives and reference optimization settings are described in Sec. 5.

Data scale and coverage. The selected data sources for main training contain 159.66M generation samples and 213.39M understanding samples. The latter comprise 99.98M captioning samples, 97.90M visual knowledge QA samples, 9.99M OCR samples, 4.51M grounding samples, and 1.00M GUI samples. These sources also include 0.50M editing examples and 624.8K interleaved sequences. This coverage supports supervision for both general semantics and specialized visual skills, while interleaved examples retain transitions between images. The total understanding count sums the five listed task groups; it is not an independently deduplicated image count. Cross-group overlap and actual use during training must be counted separately. The video construction source list in Sec. 4.4 is an upstream source pool and is not added to these listed training-source counts.

Fig. 6 pairs source dataset sizes with training examples from all nine task groups. Image generation, captioning, and visual knowledge QA account for the largest counted source pools. Editing examples preserve source–target relations, while interleaved examples retain an ordered sequence of frames. These examples illustrate task structure rather than model-generated evaluation outputs. The composition chart excludes text-only data because their total count is unavailable.

Beyond these main-training sources, the data for training video capabilities comprise 24.24M generation samples and 3.29M editing samples. Figure 7 summarizes their composition and shows representative examples. VCG accounts for 89.8% of the video generation samples, complemented by Aishi and other sources. The video editing data cover global edits (2.07M), local edits (557.4K), and simulation-to-real (Sim2Real) transformations (669.5K), with temporally aligned source–target clips. These data support exploratory post-SFT fine-tuning; their collection and filtering are described in Section 4.6.

4.1 Multimodal Understanding

Data sources and task coverage. The understanding mixture comprises image captioning, knowledge-oriented visual question answering, OCR, GUI understanding, and visual grounding. These correspond to separate sampling groups, allowing broad semantic supervision and specialized visual tasks to be balanced independently. Captioning supplies descriptions of image content, while visual questions connect image evidence to a requested answer. The specialized groups cover complementary aspects of visual understanding:

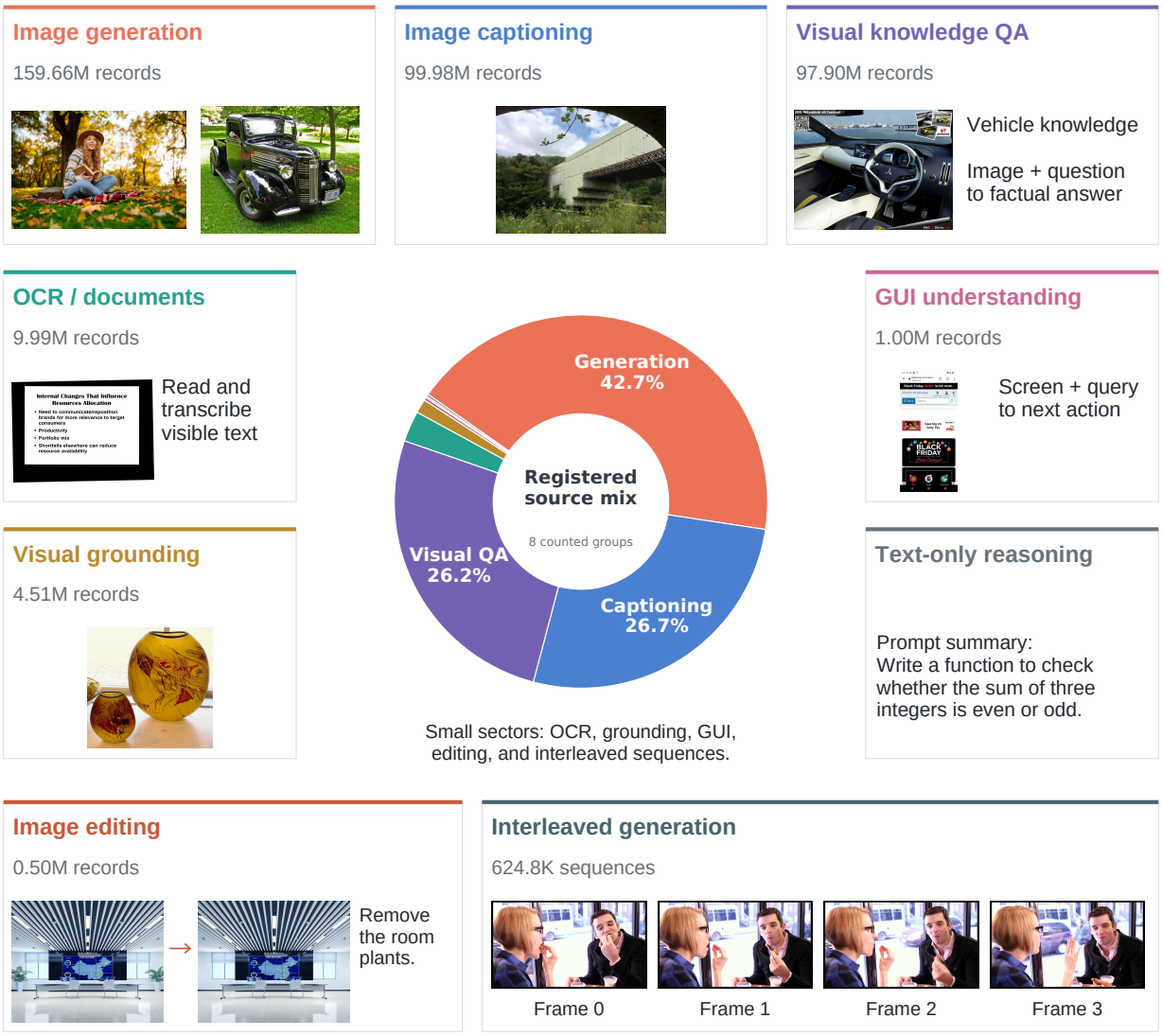


Figure 6: **Training-data composition and task examples.** Source counts and examples summarize the task coverage of joint training, from multimodal understanding to image generation, editing, and interleaved prediction.

- **OCR and document content.** Image-conditioned text supervision emphasizes reading visible text and relating it to the surrounding visual context.
- **GUI understanding.** Interface images and associated instructions connect language to visual elements in structured screens.
- **Visual grounding.** Localization-oriented examples connect referring expressions to spatially specified visual content.
- **Visual knowledge and question answering.** Image-grounded questions and responses broaden semantic coverage beyond captioning.

Input processing. Images are processed by the Qwen3.5 understanding frontend with aspect-ratio-aware resizing, patch/merge alignment, and a pixel budget. The lower-resolution mixtures use a target long-edge setting of 384; expanded mixtures use a target setting of 980, aligned to 960 by the spatial factor of 32. These transforms are independent of the generation resolution. Conversation roles and supervision masks identify the assistant text to predict, while the image remains an input condition.

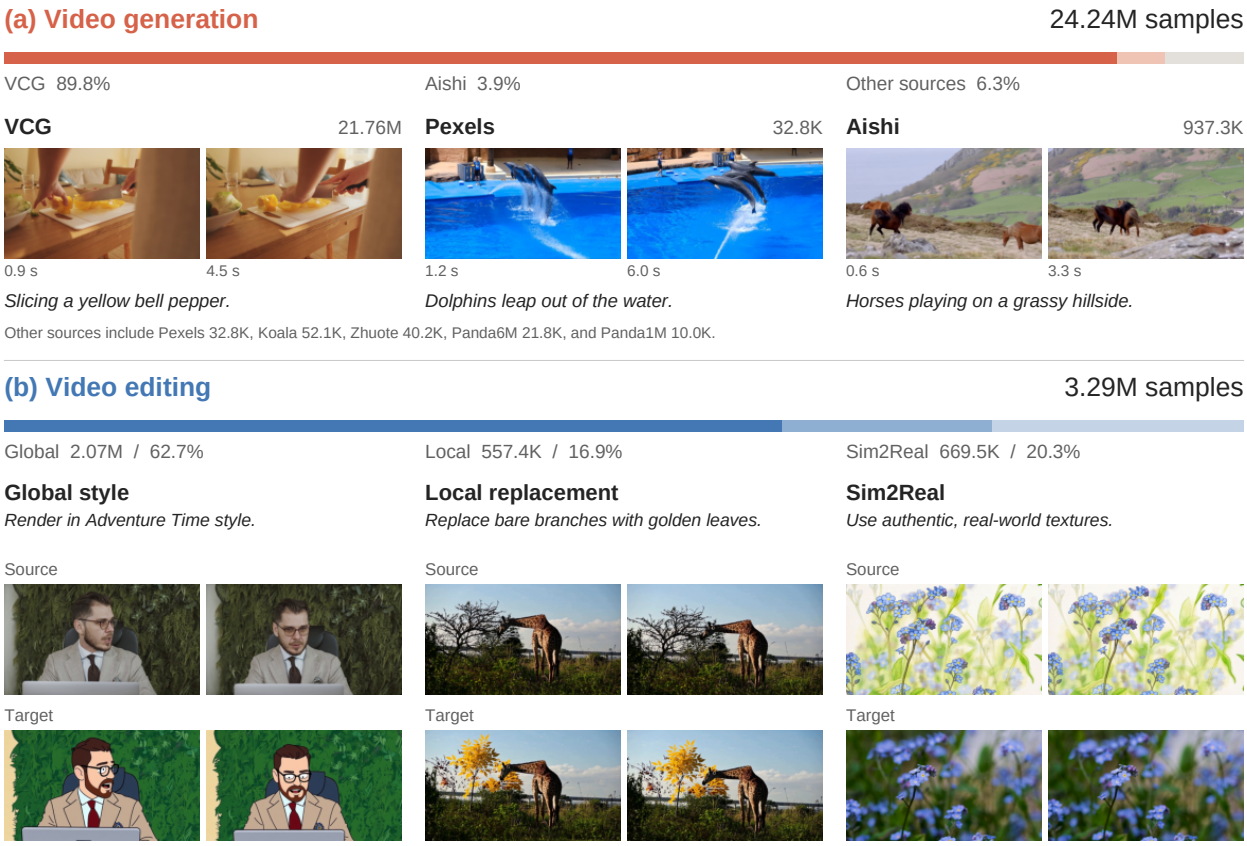


Figure 7: **Video-data composition and examples.** Generation sources (a) and aligned editing pairs (b) provide 24.24M and 3.29M samples, respectively, for post-SFT video adaptation. Counts refer to indexed samples; byte-identical Ditto shards are counted once.

SFT emphasis. Understanding SFT is organized around answer correctness, visual grounding, and instruction following. The intended selection criteria favor answers supported by the image, accurate transcription of visible text, and responses that address the requested task. Captioning, visual questions, OCR, GUI tasks, and grounding remain distinct supervision categories so that improving one capability need not remove supervision for the others.

4.2 Image Generation

Data source and representation. The generation stream provides images and associated text conditions through a dedicated image-data reader. Each example is converted into conditioning text and a target VAE latent sequence. The target receives flow-matching supervision, while text provides its semantic condition. Generation is sampled separately from understanding and editing, enabling the mixture to control their relative training exposure.

Progressive spatial preparation. Generation data follow the resolution curriculum rather than a single fixed-size preprocessing rule. The foundation setting uses 256-scale images; resolution scaling introduces a 512-scale setting; high-resolution refinement extends the range through 1024-scale settings and up to 2K. Images retain their aspect-ratio structure subject to resize, stride, and pixel-budget constraints, so image dimensions vary within each scale. Semantic image inputs retain their own understanding-side transform.

SFT emphasis. Generation SFT targets agreement between the instruction and the image, including the requested subjects, attributes, spatial relations, and visible text. Its data selection should emphasize reliable image–text correspondence and visual fidelity, rather than increasing resolution alone. The intended supervi-

sion includes demanding compositions and precise instructions, with selection guided by the errors observed during generation evaluation.

4.3 Image Editing

Data source and processing. The editing stream contains an instruction, a source image, and a target image. The source is retained as public visual context through semantic features and, where configured, clean VAE latents. The target is represented as a private noisy latent block carrying the visual loss. The sequence plan keeps the source and target roles explicit, preserving the direction of the transformation. Generation-side resize settings apply to editing images, while semantic reference features use the understanding transform.

Instruction and target consistency. The central SFT criterion for editing is correspondence between the stated change and the source–target pair. Intended quality checks distinguish successful edits from unchanged targets or visible artifacts, and examine whether unrelated content remains consistent. Precise edit instructions should describe the actual transformation without adding unsupported changes. This criterion targets both instruction following and preservation of reference content.

4.4 Interleaved Data

Sources and sequence construction. The interleaved stream builds related text–image sequences from seven listed visual-sequence collections. Its adapters support alternating text–image examples and chained image edits. The interleave-enabled joint mixtures use the chained-edit format, retaining transition text and target captions. Each chain begins with an opening prompt and source image, then specifies successive transitions and visual targets:

$$(P, I_0) \longrightarrow [\Delta_1, D_1, I_1, \Delta_2, D_2, I_2, \ldots], \tag{9}$$

where Δ_k is a transition instruction, D_k is the target caption, and I_k is the corresponding target image. Reasoning fields used during data construction are omitted from the training response.

Video sources and annotation settings. Our video annotation effort draws on existing text-to-video, egocentric, and interaction-rich video collections. The batch annotation source list contains 24,235,299 source samples (Tab. 1). This source list is separate from the seven listed training collections above: it measures available inputs to data construction, before filtering, annotation acceptance, or training-mixture selection. OpenVid is also listed as an input, but its sample count is unavailable and is excluded from the total.

Table 1: **Video annotation sources.** Source samples before filtering and annotation acceptance.

Source identifier	Source samples
Panda1M	10,023
Panda6M	21,823
Koala clips	52,081
Editing	1,377,059
VCG	21,764,019
Pexel	32,791
Aishi	937,318
Zhuote	40,185
Total	24,235,299

Event-driven keyframes and transition annotation. The construction protocol asks a video-language model to select three to five timestamps marking meaningful visual changes, such as contact, grasping, release, object movement, or a camera-view change. Timestamps must be strictly increasing; the first is normally zero unless the opening frames are invalid. The batch configuration samples video observations at 1 FPS. This sampling rate controls the annotation input, whereas the selected keyframes follow event boundaries rather than a fixed output interval. The annotation model supplies descriptions and transition instructions for the selected frames.

The first instruction describes the initial static scene. Subsequent instructions combine a target-state description D_k with a transition Δ_k from the preceding keyframe. The protocol emphasizes subject pose, object position and orientation, contact relations, and changes in scene or camera viewpoint, including contact changes, object motion, and final placement. Annotation must remain grounded in visible evidence rather than inferred intentions or unobserved outcomes. Timestamp references are resolved to extracted frames, producing alternating user-text and assistant-image messages. This yields Eq. (9) after adaptation to the chained-edit format. The visual targets are observed video frames; the annotation model supplies their descriptions and transition instructions.

Quality screening and annotation validation. The planned screening uses resolution, duration, clarity, exposure, aesthetics, motion, shot boundaries, and overlays to identify unsuitable clips and split clips longer than 30 seconds. Source counts are reported before this screening.

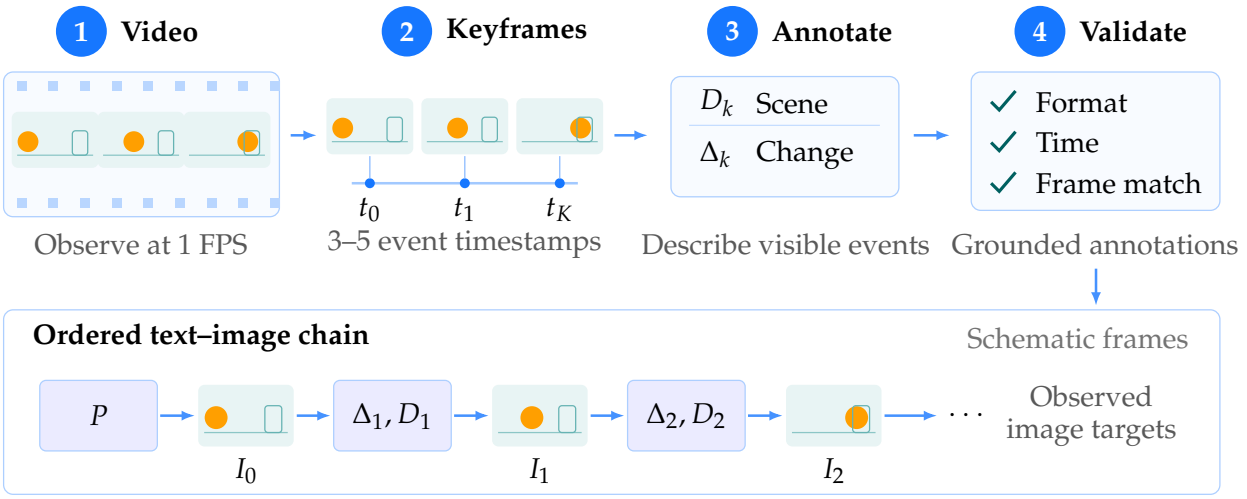


Figure 8: **Video-based interleaved data construction.** Event-selected video frames are paired with scene descriptions and transition instructions to form grounded text–image training sequences.

Before use, annotations must be checked for correct format and timing: valid role alternation, one target image per assistant turn, valid media references, strictly increasing in-range timestamps, and consistency between transition text and the adjacent frames. The image-target protocol keeps visual-state descriptions and transition instructions tied to the extracted frames.

Annotation pipeline. Video decoding, quality screening, and annotation are separate steps, allowing source-specific filtering while keeping a common output format.

Structural filtering and clean context. Samples must be parsed successfully and checked for suitability for interleaved training. The adapter also verifies the message structure and requires each successive source image reference to match the preceding target reference. Non-final target images are followed by clean representations of the observed image, making completed visual content available to later text and image prediction. Clean ViT features provide semantic context; clean VAE context is retained according to the packing configuration. These teacher-forced context copies carry no visual denoising loss. The target and context share a semantic image identity while following the private/public information-flow rules in Sec. 3.4.

SFT emphasis and stage coverage. Interleaved SFT is intended to improve transitions between outputs, consistency with preceding images, and the ability to follow sequential instructions. Examples should be selected as complete coherent sequences rather than independent image–text pairs. Interleaving is enabled in selected joint training mixtures, but not every high-resolution configuration retains it. The final SFT curriculum therefore treats interleaved supervision as an explicit mixture choice.

4.5 Text-Only Data

Language and reasoning supervision. The text-only group lists 100 source entries covering mathematics, programming, knowledge, reasoning, general instructions, conversation, tool use, and long-context tasks. It supplies direct language supervision alongside visual learning. Reasoning examples belong to this text mixture; they are distinct from the reasoning used to construct interleaved editing samples, which is removed before training. During capability refinement, the intended role of text data is to maintain response quality and instruction-following coverage while understanding and generation are refined jointly.

4.6 Video Data

The video data comprise separate groups for video generation and source-conditioned video editing, used in the exploratory post-SFT fine-tuning described above. Video generation data are collected from VCG, Koala, Aishi, Pexels, Panda, and internal sources, while video editing data are drawn from Ditto (Bai et al., 2026). We exclude videos longer than eight seconds and those with excessively low or high original frame rates. We further filter the data using VLM-based assessments of visible text, aesthetics, and visual quality. Retained videos are resampled to 16 FPS; for editing, source and target clips share the same temporal sampling plan.

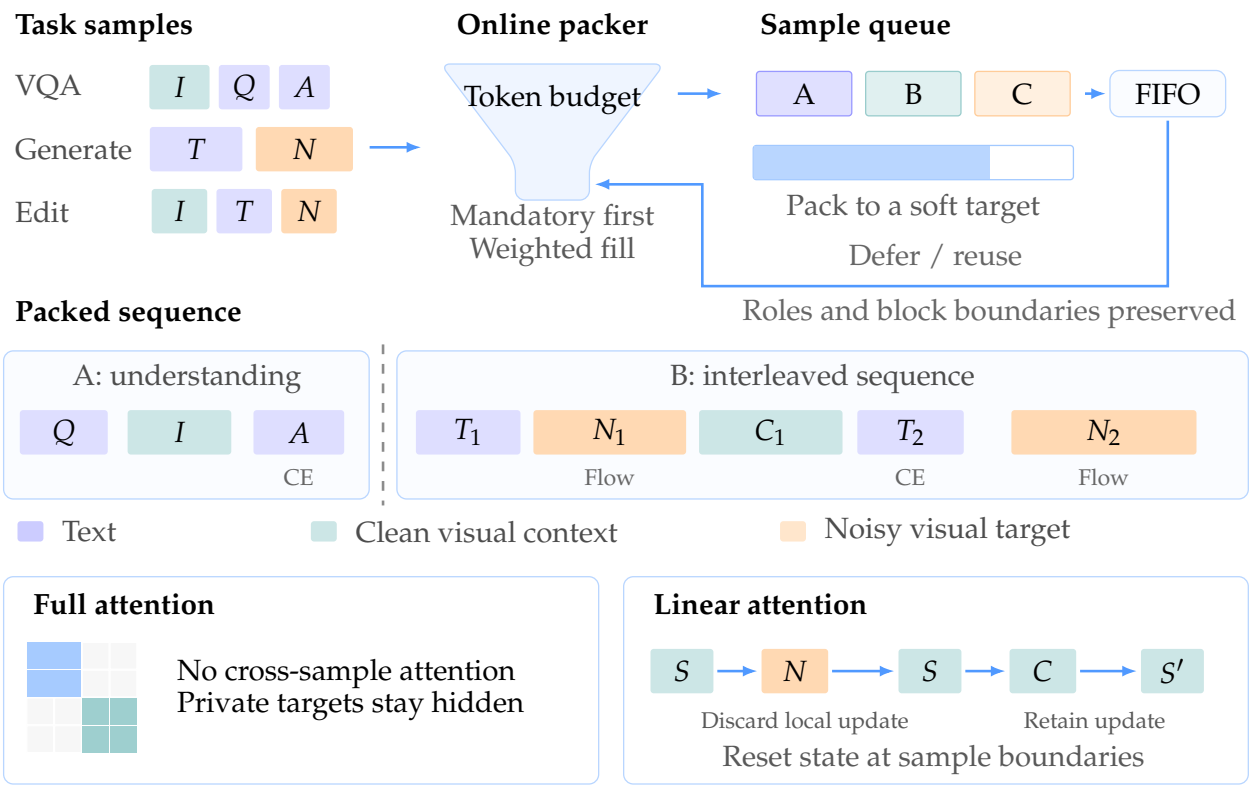


Figure 9: **Multimodal sequence packing.** Online packing combines task-specific samples within a token budget while preserving sample isolation and the public/private boundaries required by hybrid attention.

4.7 Multimodal Sequence Packing

Fig. 9 illustrates how task-specific samples become a packed sequence while retaining the boundaries needed by hybrid attention and the two prediction objectives.

Unified sequence assembly. The task adapters convert samples from different tasks into a common sequence plan containing text roles, media identities, attention-block types, and supervision locations. Packing concatenates multiple examples while retaining their individual sequence boundaries and within-example block structure. Text labels and visual targets are gathered through separate loss indices, so conditioning content can share the input sequence without receiving a prediction loss. For interleaved samples, the complete sequence of transitions and images remains within the same example; packing does not turn successive editing steps into unrelated image–text pairs.

Online token-budget packing. The loader starts each pack by inserting eligible examples from mandatory task groups. It then selects additional examples according to normalized group weights, using modality-aware token estimates to account for text and visual content. A soft token target determines when a pack is normally produced, while a per-example threshold rejects oversized samples and a separate packed-token budget guides additional insertions. Examples that do not fit can be stored in a fixed-size buffer for later use. The loader reuses buffered examples in first-in, first-out order when the current pack is below a configured occupancy threshold. This online policy accommodates the changing token costs of image resolution, aspect ratio, and interleaved sequence length without requiring a fixed number of images per batch. The final accepted example or mandatory insertion can exceed the nominal token budget. Reference budgets are listed in Tab. 3.

Sample isolation and hybrid-attention metadata. Packed sample lengths prevent attention or recurrent state from crossing between independent examples. Within an example, block lengths and attention modes preserve the conditioning relationships described in Sec. 3.2. In particular, a noisy visual target remains private, whereas an observed clean image may contribute public context for subsequent prediction. The same sequence layout supplies visibility metadata to full-attention layers and state-retention boundaries to linear-attention layers. These layers retain their respective attention behavior. Variable-length execution

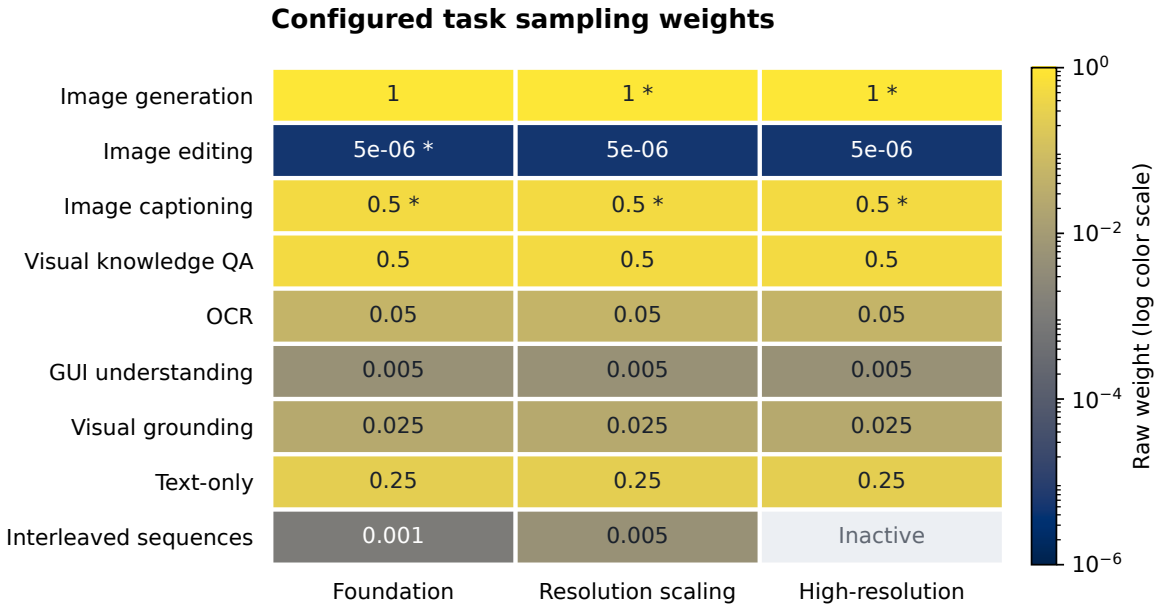


Figure 10: **Data sampling across training stages.** Representative task weights are shown for three pre-SFT stages; asterisks mark mandatory groups. These weights do not represent observed sample or token proportions.

Stage	Name	Generation scale	Primary objective
I	Joint Multimodal Foundation	256-scale	Establish joint text, visual understanding, and latent-generation modeling.
II	Progressive Resolution Scaling	512-scale	Expand spatial detail while continuing multimodal supervision.
III	High-Resolution Refinement	1024-scale to 2K	Extend visual generation to longer target sequences and finer detail.
IV	Supervised Capability Refinement (SFT)	Task-dependent	Refine understanding accuracy, instruction following, generation quality, and editing consistency.

Table 2: **Four-stage training curriculum.** Resolution labels describe training scales rather than fixed square image sizes.

can avoid padding every example to a common text–visual sequence length; visual tensor padding and backend-specific alignment can still be required.

Data mixture statistics. Group weights govern sampling decisions, rather than final sample, token, or compute percentages. Mandatory insertion and buffering further affect the actual mixture. We therefore distinguish configured weights from consumed examples and tokens when describing data allocation across stages. SFT uses the same packing interfaces, with task selection and weighting focused on capability refinement.

Fig. 10 compares sampling weights across three training stages. Required task groups are included in each pack regardless of their sampling weight. These weights do not directly determine the sample or token proportions used during training. Colors encode weights on a logarithmic scale; asterisks identify required groups and gray cells identify inactive groups. The high-resolution example uses a 1024-scale configuration.

5 Training Strategy

We organize the UniSpark curriculum by its modeling objectives: *Joint Multimodal Foundation*, *Progressive Resolution Scaling*, *High-Resolution Refinement*, and *Supervised Capability Refinement (SFT)*. This organization separates learning a shared multimodal backbone, increasing spatial detail, extending high-resolution generation, and refining task performance. Tab. 2 summarizes the four stages. The backbone retains the joint

Parameter	Foundation	Resolution scaling	High-resolution[a]	SFT[b]
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}
LR schedule	Constant	Constant	Constant	Constant
Warmup updates	2,500	2,500	2,500	500
EMA decay	0.9999	0.9999	0.9999	0.995
Timestep shift	1.0	1.0	1.0	4.0
Text / visual loss weights	1 / 1	1 / 1	0.25 / 1	0.25 / 1
Packed-token target	16,384	10,240	16,384	10,240
Per-sample token limit	26,384	16,384	26,384	16,384
Packed-token limit	46,864	36,864	46,864	36,864

Table 3: **Reference optimization settings by training stage.** The packed-token target applies to one packed sequence, not a global batch. [a]High-resolution values refer to the 1024-scale setting. [b]SFT settings are preliminary. All learning-rate schedules are constant after warmup.

text and visual objectives in Sec. 3.3 throughout this design.

5.1 Joint Multimodal Foundation

The foundation stage trains the hybrid MoT backbone with a low visual token budget. Text and semantic visual tokens use the understanding expert, while VAE latent tokens use the generation expert. Language modeling and visual flow matching provide direct supervision for the two output types. Image generation, image editing, understanding, and text examples enter the common packed-sequence format, with optional interleaved supervision determined by the data mixture.

The 256-scale setting keeps visual targets relatively short while establishing interactions between the two experts. The language backbone, vision encoder, and expert branches and adapters are trained jointly, while the VAE remains frozen. The stage's purpose is to establish the shared prediction interface before increasing the spatial sequence length.

5.2 Progressive Resolution Scaling

The resolution-scaling stage increases generation to a 512-scale setting while retaining the same token roles and loss definitions. More latent patches represent each image, allowing training to cover finer spatial detail. Understanding inputs have an independent resolution budget, so higher-resolution generation does not require every semantic condition to expand at the same rate.

Interleaved sequences can continue to contribute text and visual targets in this stage. Clean representations of earlier images remain public context, while noisy prediction blocks remain private. This exposes hybrid attention to both longer visual blocks and dependencies across multiple outputs. Changes in resolution and task sampling are therefore tracked separately.

5.3 High-Resolution Refinement

High-resolution refinement extends generation through 1024-scale settings and up to 2K. The objective remains conditional flow matching, with understanding and language supervision retained in the joint training design. The increased latent-token count makes memory use and packed sequence composition more sensitive to aspect ratio and the number of visual targets per example. Token budgets and task weights can therefore change without changing the underlying architecture.

The 1024-scale reference setting assigns coefficients of 0.25 and 1.0 to the text and visual losses, respectively. These are loss multipliers, not a text-to-image data ratio. It also omits the interleaved data group, whereas a preceding resolution-scaling configuration includes that group. High-resolution refinement thus need not preserve every task at the same sampling rate. Extending the stage to 2K retains the distinction between generation resolution, understanding resolution, and the number of samples that fit in a packed sequence.

5.4 Supervised Capability Refinement

SFT is the final capability-oriented stage of the main multimodal training curriculum. Its goal is to improve understanding and generation together through task-focused supervision, using the data categories in Sec. 4. The refinement design has four priorities:

- **Understanding accuracy:** answer visual questions, recognize text, and ground responses in the supplied

image.
- **Generation fidelity:** follow the requested subjects, attributes, spatial relations, and text in the image.
- **Editing consistency:** perform the instructed change while retaining unrelated reference content.
- **Interleaved coherence:** use preceding visual outputs and text consistently when predicting subsequent content.

The same text cross-entropy and visual flow-matching objectives support these tasks. SFT changes the emphasis of data selection and sampling; it does not require a separate understanding model or visual generator.

Tab. 3 gives preliminary SFT settings; the final data mixture and schedule are not fixed.

5.5 Optimization

Optimizer and precision. Training uses AdamW (Loshchilov & Hutter, 2019) with coefficients $(\beta_1, \beta_2) = (0.9, 0.95)$, numerical stability constant 10^{-15}, zero weight decay, and gradient-norm clipping at 1.0. Forward computation uses bfloat16 mixed precision. The joint-training reference settings use learning rate 10^{-4} with a constant schedule after 2,500 warmup updates and EMA decay 0.9999. Tab. 3 summarizes the reference settings for all four stages.

Noise-time sampling. The visual objective samples a time through a sigmoid transform and then applies the configured shift s:

$$t' = \frac{s\,t}{1 + (s-1)t}. \tag{10}$$

The joint configurations set $s = 1$, while the SFT reference uses $s = 4$. The latter gives larger noise times for intermediate t under the convention of Eq. (5). This parameter changes the training time distribution, not the number of denoising steps used at inference.

5.6 Adaptation to Video Generation and Editing

Following SFT, we further fine-tune the model on video generation and editing data as an exploratory extension of its visual capabilities. The fine-tuning mixture also retains image generation and editing data. Video generation, video editing, image generation, and image editing are sampled at a ratio of $5 : 1 : 1 : 0.1$. For video generation and editing, the longer side of each frame is set to 1280 and 768 pixels, respectively, while image generation and editing retain 2K resolution. Video clips are sampled at 16 FPS and span 3–5.2 seconds.

The understanding branch is frozen during this adaptation. Tab. 4 summarizes the fine-tuning settings.

Parameter	Value
Learning rate	5×10^{-5}
LR schedule	Constant
Warmup updates	1,000
EMA decay	0.999
Timestep shift	1.0
Text conditioning dropout	0.1
VAE / ViT conditioning dropout	0.3 / 0.3
Packed-token target	16,384
Per-sample token limit	40,960
Packed-token limit	40,960

Table 4: **Optimization settings for video adaptation stage.** The understanding branch is frozen.

5.7 Training Infrastructure and Efficiency

Hybrid sharding and activation memory. Training uses FSDP (Zhao et al., 2023) hybrid sharding to distribute parameters, gradients, and optimizer states across GPUs. Decoder-layer and vision-encoder activation checkpointing reduce memory use through recomputation (Chen et al., 2016). These methods support training with longer visual sequences.

Attention execution for packed multimodal sequences. Variable-length attention backends directly use the packing metadata in Sec. 4.7. The full-attention implementation supports block-mask execution, caches derived block-sparse metadata, and provides a native grouped-query path that avoids explicitly repeating key/value heads. Backend selection depends on the head configuration, requested execution path, and installed kernels. The linear-attention MoT implementation provides optimized execution paths alongside a scalar reference implementation. Expert-specific projections and convolution weights operate with shared public recurrent state and convolution history. Private target blocks read this context but do not retain their local updates in public context. Optimizing these operations must preserve both sample isolation and the public/private state rules; full-attention visibility and linear-attention recurrence are not interchangeable.

Data delivery and visual encoding. The input pipeline prefetches samples and assembles packed batches. A frozen VAE encodes visual targets and conditions before the backbone forward pass, reducing the memory required for visual encoding.

Workload-aware monitoring. The implementation supports text and visual loss monitoring by task group, records token throughput, and estimates utilization with an architecture-aware FLOP model that distinguishes full-attention and linear-attention layers and accounts for visual computation. Executed-compute estimates also consider activation recomputation. Token throughput, peak memory, and packed sequence lengths provide complementary workload measurements.

6 Experiments

We evaluate UniSpark on multimodal understanding, text-to-image generation, and instruction-based image editing, and measure inference latency and memory across image resolutions. The comparisons include both task-specific models and unified multimodal models. Fig. 1 summarizes selected benchmarks, while the following tables provide per-benchmark and per-category results.

6.1 Evaluation Setup and Baselines

UniSpark-9B uses approximately 9B parameters in each of its understanding and generation branches. Model-size conventions and inference settings are specified in the corresponding comparisons.

For multimodal understanding, we use MME (Fu et al., 2023) for perception and cognition, MMBench (Liu et al., 2024) and MM-Vet (Yu et al., 2024) for visual question answering and integrated multimodal capabilities, MMMU (Yue et al., 2024) for subject-level reasoning, MathVista (Lu et al., 2023) for visual mathematics, and MMVP (Tong et al., 2024b) for fine-grained visual discrimination.

For text-to-image generation, we evaluate compositional alignment with GenEval (Ghosh et al., 2023) and detailed prompt following with DPG-Bench (Hu et al., 2024). For instruction-based image editing, ImgEdit (Ye et al., 2025) measures whether the model makes the requested changes while preserving unrelated content. The following subsections specify evaluation protocols and analyze the corresponding results.

We compare understanding-only models, including InternVL2/2.5 (OpenGVLab, 2024; Chen et al., 2024) and Qwen2/2.5-VL (Wang et al., 2024b; Bai et al., 2025) families, Kimi-VL (Kimi Team et al., 2025a), and DeepSeek-VL2 (Wu et al., 2024c), with unified models such as Janus-Pro (Chen et al., 2025), MetaQuery-XL (Pan et al., 2025), BAGEL (Deng et al., 2025), and SenseNova-U1 (Diao et al., 2026). Qwen3.5-4B and Qwen3.5-9B (Qwen Team, 2026) provide additional understanding references. This grouping compares UniSpark with both specialized vision-language models and models that also generate images.

The GenEval and DPG-Bench comparisons include specialized generators such as FLUX.1 [Dev] (Black Forest Labs, 2024), Seedream 3.0 (Gao et al., 2025), Qwen-Image (Wu et al., 2025a), Emu3-Gen (Wang et al., 2024c), and Lumina-mGPT 2.0 (Xin et al., 2025b). Unified baselines include BAGEL, OmniGen2 (Wu et al., 2025b), InternVL-U (Tian et al., 2026), Lumina-DiMOO (Xin et al., 2025a), and LLaDA2.0-Uni (Inclusion AI et al., 2026). The tables distinguish autoregressive, continuous-diffusion, and discrete-diffusion architectures.

ImgEdit compares four specialized editors (FLUX.1 Kontext (Black Forest Labs et al., 2025), Step1X-Edit (Liu et al., 2025), Qwen-Image-Edit (Qwen Team, 2025), and Z-Image-Edit (Z-Image Team et al., 2025)) with five unified models: BAGEL, OmniGen2, InternVL-U, Lumina-DiMOO, and LLaDA2.0-Uni. For inference efficiency, BAGEL and SenseNova-U1 serve as references under matched hardware and workload settings.

Understanding baseline scores come from BAGEL's comparison and additional Qwen3.5 and SenseNova-U1 evaluations. Generation and editing baselines follow LLaDA2.0-Uni (Inclusion AI et al., 2026). Published scores retain their source evaluation settings; UniSpark scores and the inference-efficiency measurements are reported separately. Dashes indicate unreported results.

6.2 Multimodal Understanding

We report MME perception (MME-P) and combined perception-plus-cognition (MME-S), MMBench v1.0-dev-en circular accuracy, MMMU validation accuracy without CoT, and MMVP paired accuracy. Protocol differences among the reported models are documented in Tab. 5.

In Tab. 5, UniSpark records the highest listed MME-P (1766.0), MME-S (2470.3), MMBench (86.7), and MMVP (72.0) scores, with the MMBench protocol differences noted in the caption. Compared with BAGEL (7B MoT), it scores higher across all six benchmark families, including gains of 6.2 points on MM-Vet and 4.1 points

Type	Model	# LLM Params	MME-P↑	MME-S↑	MMBench↑	MMMU↑	MM-Vet↑	MathVista↑	MMVP↑
Und. Only	InternVL2 (2024)	1.8B	1440.0	1877.0	73.2	34.3	44.6	46.4	35.3
	InternVL2.5 (2024)	1.8B	–	2138.0	74.7	43.6	60.8	51.3	–
	Qwen2-VL (2024b)	1.5B	–	1872.0	74.9	41.1	49.5	43.0	–
	BLIP-3 (2024)	4B	–	–	76.8	41.1	–	39.6	–
	LLaVA-OV (2024)	7B	1580.0	–	80.8	48.8	57.5	63.2	–
	InternVL2 (2024)	7B	1648.0	2210.0	81.7	49.3	54.2	58.3	51.3
	InternVL2.5 (2024)	7B	–	2344.0	84.6	56.0	62.8	64.4	–
	Qwen2-VL (2024b)	7B	–	2327.0	83.0	54.1	62.0	58.2	–
	Emu3-Chat** (2024c)	8B	1244.0	–	58.5	31.6	37.2	–	36.6
	DeepSeek-VL2 (2024c)	4.1B/28B	–	–	–	51.1	60.0	62.8	–
	Qwen2.5-VL (2025)	3B	–	2157.0	79.1	53.1	61.8	62.3	–
	Qwen2.5-VL (2025)	7B	–	2347.0	83.5	**58.6**	67.1	68.2	–
	Kimi-VL (2025a)	2.8B/16B	–	–	–	57.0	66.7	68.7	–
	Qwen3.5-4B (2026)	4B	1686.5	2348.3	81.4	29.2	76.5	**80.7**	64.7
	Qwen3.5-9B (2026)	9B	1748.7	2439.1	85.1	43.1	**79.6**	79.6	70.7
Unified	Show-o$_{512}$ (2024a)	1.3B	1097.0	–	–	26.7	–	–	–
	Janus (2024a)	1.5B	1338.0	–	69.4	30.5	34.3	–	–
	ILLUME (2024a)	7B	1445.0	–	75.1	38.2	37.0	–	–
	VILA-U$^{**}_{256}$ (2024b)	7B	1336.0	–	66.6	32.2	27.7	–	22.0
	Chameleon** (2024)	7B	–	–	35.7	28.4	8.3	–	0.0
	LlamaFusion** (2024)	8B	1604.0	–	72.1	41.7	–	–	–
	MetaMorph (2024a)	8B	–	–	75.2	41.8	–	–	48.3
	SEED-X (2024)	13B	1457.0	–	70.1	35.6	43.0	–	–
	TokenFlow-XL (2024)	13B	1546.0	–	68.9	38.7	40.7	–	–
	MUSE-VL (2024b)	32B	–	–	81.8	50.1	–	55.9	–
	Janus-Pro (2025)	1.5B	1444.0	–	75.5	36.3	39.8	–	–
	BAGEL (2025)	1.5B MoT	1610.0	2183.0	79.2	43.2	48.2	63.4	54.7
	Janus-Pro (2025)	7B	1567.0	–	79.2	41.0	50.0	–	–
	MetaQuery-XL† (2025)	7B	1685.0	–	83.5	**58.6**	66.6	–	–
	BAGEL (2025)	7B MoT	1687.0	2388.0	85.0	55.3	67.2	73.1	69.3
	SenseNova-U1 (2026)	8B MoT	1683.5	2343.5	82.8	53.1	63.0	71.5	58.7
	UniSpark	9B MoT	**1766.0**	**2470.3**	**86.7**	56.7	73.4	77.2	**72.0**

Table 5: **Bold** and underlining indicate the best and second-best scores. MoT parameters are per branch; MoE parameters are active/total. MMBench answer extraction differs: UniSpark uses rules only, while Qwen3.5 and SenseNova-U1 use a Qwen fallback. †: frozen Qwen2.5-VL backbone. **: scores taken from MetaMorph or MetaQuery in the source comparison. Dashes indicate unreported scores.

on MathVista. These results indicate broad understanding capability spanning perception, visual question answering, and visual mathematics.

The remaining gaps are concentrated in specific reasoning benchmarks. On MMMU-Val, UniSpark scores 56.7, below MetaQuery-XL's 58.6. Qwen3.5-9B also scores higher on MM-Vet and MathVista by 6.2 and 2.4 points, respectively. Thus, the strong perception results do not imply uniformly better reasoning. Differences in model size, training data, and evaluation protocols also prevent attributing these comparisons solely to hybrid attention.

6.3 Text-to-Image Generation

We report overall and category-level GenEval and DPG-Bench scores.

Baseline scores in Tab. 6 and Tab. 7 follow LLaDA2.0-Uni (Inclusion AI et al., 2026). We report UniSpark results after unified supervised fine-tuning (SFT). We label Lumina-mGPT 2.0 as autoregressive (AR), following its architecture (Xin et al., 2025b); this corrects the diffusion label in the DPG-Bench source table.

After unified SFT, UniSpark achieves an overall GenEval score of 0.83 (Tab. 6), close to BAGEL's direct-generation score of 0.82. Its scores are highest on single objects (0.99), colors (0.94), and two objects (0.93), while counting (0.68), position (0.67), and attribute binding (0.73) remain more difficult. This pattern shows stronger object and color fidelity than precise composition. LLaDA2.0-Uni and Lumina-DiMOO reach higher

Type	Model	Architecture	Single object	Two objects	Counting	Colors	Position	Attribute binding	Overall↑
Gen. Only	FLUX.1 [Dev] (2024)	CD	0.98	0.81	0.74	0.79	0.22	0.45	0.66
	Emu3-Gen (2024c)	AR	0.98	0.71	0.34	0.81	0.17	0.21	0.54
	Lumina-mGPT 2.0 (2025b)	AR	0.99	0.87	0.44	0.85	0.44	0.54	0.69
	Seedream 3.0 (2025)	CD	0.99	0.96	0.91	0.93	0.47	0.80	0.84
	Qwen-Image (2025a)	CD	0.99	0.92	0.89	0.88	0.76	0.77	0.87
	LongCat-Image (2025)	CD	0.99	0.98	0.86	0.86	0.75	0.73	0.87
	Z-Image-Turbo (2025)	CD	1.00	0.95	0.77	0.89	0.65	0.68	0.82
Unified	Janus-Pro (2025)	AR	0.99	0.89	0.59	0.90	0.79	0.66	0.80
	BAGEL (2025)	AR + CD	0.99	0.94	0.81	0.88	0.64	0.63	0.82
	OmniGen2 (2025b)	AR + CD	1.00	0.95	0.64	0.88	0.55	0.76	0.80
	HunyuanImage-3.0 (2025)	AR + CD	1.00	0.92	0.48	0.82	0.42	0.63	0.72
	MMaDA (2025a)	DD	0.99	0.76	0.61	0.84	0.20	0.37	0.63
	Lumina-DiMOO (2025a)	DD	1.00	0.94	0.85	0.89	0.85	0.76	0.88
	NextFlow (2026)	AR + CD	0.98	0.92	0.73	0.90	0.77	0.69	0.83
	InternVL-U (2026)	AR + CD	0.99	0.94	0.74	0.91	0.77	0.74	0.85
	LLaDA-o (2026)	DD + CD	0.99	0.98	0.73	0.96	0.69	0.83	0.86
	LLaDA2.0-Uni (2026)	DD + CD	1.00	0.98	0.73	0.92	0.90	0.84	0.89
	UniSpark	AR + CD	0.99	0.93	0.68	0.94	0.67	0.73	0.83

Table 6: **Text-to-image generation on GenEval.** Scores range from 0 to 1; higher is better. The blue row reports UniSpark after unified SFT. AR, CD, and DD denote autoregressive, continuous diffusion, and discrete diffusion models, respectively; "+" indicates combined architectures.

overall scores of 0.89 and 0.88, leaving room to improve compositional control. The BAGEL comparison uses direct generation, keeping results with external prompt rewriting separate.

Tab. 7 reports a DPG-Bench overall score of 80.54 for UniSpark, below BAGEL (85.07) and LLaDA2.0-Uni (87.76). Within its category results, Entity is strongest at 82.62, whereas Attribute (77.07), Relation (76.71), and Other (73.50) are lower. Together with GenEval, this pattern points to attribute and relational control as priorities for improving prompt adherence.

6.4 Instruction-Based Image Editing

We evaluate UniSpark on all 737 ImgEdit samples using Qwen3-VL-235B-A22B-Thinking as the judge. Tab. 8 reports nine editing categories and the overall score. Baseline scores follow LLaDA2.0-Uni (Inclusion AI et al., 2026). Because the judges differ, our scores are reference results and are not directly comparable to those baselines.

In Tab. 8, UniSpark obtains an overall reference score of 3.13. Style editing is its strongest category (4.05), followed by object addition (3.72). Extraction (1.68) and hybrid editing (1.62) receive the lowest scores, identifying these operations as the main weaknesses under the current evaluator. This variation shows why the overall score alone is insufficient to characterize editing ability. Because our Qwen3-VL judge differs from the judge used for the published baselines, these scores support analysis across our editing categories but do not establish a direct ranking against other models.

6.5 Interleaved Generation and Sequence Editing

Text-only generation jointly produces F1–F4. Sequence editing takes the same F1 as input and jointly generates F2–F4.

Fig. 11 shows local changes in ribbon curvature, jellyfish tentacles, and flame shape with recognizable subjects and backgrounds. Flame direction and bell contraction remain imperfect. Selected from 12 paired runs, these cases illustrate visual continuity, not reliable execution of every requested stage.

6.6 Extension to Video Generation and Editing

Following the exploratory post-SFT adaptation described in Sec. 5.6, we examine the model's video generation and editing capabilities through qualitative examples.

Fig. 12 presents text-conditioned video generation examples spanning camera movement around a building,

Type	Model	Architecture	Global	Entity	Attribute	Relation	Other	Overall↑
Gen. Only	FLUX.1 [Dev] (2024)	CD	74.35	90.00	88.96	90.87	88.33	83.84
	Emu3-Gen (2024c)	AR	85.21	86.68	86.84	90.22	83.15	80.60
	Lumina-mGPT 2.0 (2025b)	AR	86.63	91.97	90.20	94.85	84.80	87.20
	Seedream 3.0 (2025)	CD	94.31	92.65	91.36	92.78	88.24	88.27
	Qwen-Image (2025a)	CD	91.32	91.56	92.02	94.31	92.73	88.32
	LongCat-Image (2025)	CD	89.10	92.54	92.00	93.28	87.50	86.80
	Z-Image-Turbo (2025)	CD	91.29	89.59	90.14	92.16	88.68	84.86
Unified	Janus-Pro (2025)	AR	86.90	88.90	89.40	89.32	89.48	84.19
	BAGEL (2025)	AR + CD	88.94	90.37	91.29	90.82	88.67	85.07
	OmniGen2 (2025b)	AR + CD	88.81	88.83	90.18	89.37	90.27	83.57
	HunyuanImage-3.0 (2025)	AR + CD	92.12	92.53	89.13	92.13	91.92	86.10
	MMaDA (2025a)	DD	77.81	78.48	81.74	84.79	63.20	69.97
	Lumina-DiMOO (2025a)	DD	81.46	92.08	88.98	94.31	82.00	86.04
	NextFlow (2026)	AR + CD	92.40	90.05	90.51	92.72	91.14	86.00
	InternVL-U (2026)	AR + CD	90.39	90.78	90.68	90.29	88.77	85.18
	LLaDA-o (2026)	DD + CD	92.91	93.30	90.40	91.75	92.79	87.04
	LLaDA2.0-Uni (2026)	DD + CD	91.14	93.55	91.98	92.17	93.18	87.76
	UniSpark	AR + CD	79.33	82.62	77.07	76.71	73.50	80.54

Table 7: **Text-to-image generation on DPG-Bench.** Higher is better. The blue row reports UniSpark after unified SFT. AR, CD, and DD denote autoregressive, continuous diffusion, and discrete diffusion models, respectively; "+" indicates combined architectures.

waves along a shoreline, and subtle human motion. Each example includes its text prompt and five frames sampled from the generated clip, illustrating scene appearance and its evolution over time.

Fig. 13 presents source-conditioned video editing examples covering object replacement and global stylization. The first replaces a black plate with a terracotta dish, while the second transforms the video into a pencil-sketch style. Source and edited frames are shown at corresponding timestamps to illustrate the requested changes alongside the preservation of scene content and action across frames.

6.7 Inference Efficiency and Scaling

We compare UniSpark with BAGEL (Deng et al., 2025) and SenseNova-U1 (Diao et al., 2026) on image generation and visual understanding. Generation is evaluated at matched output resolutions; understanding is evaluated at matched visual-token budgets with a fixed text output length. Each model runs on a dedicated H20-3e GPU in BF16 at batch size 1, with all model components GPU-resident. Each setting uses five warm-ups and four input cases repeated three times. We report median latency and throughput, descriptive P95 generation latency, and the maximum allocated GPU memory across the 12 measured runs. Model loading and disk I/O are excluded. Parameter counts in Tables 9 and 10 follow the per-branch MoT convention used in Tab. 5.

For image generation, all models use the same prompts, 49 sampling updates, and 79 denoising forwards. Text CFG is set to 4 over the noise-level interval $(0.4, 1.0]$, with time shift 1 and CFG normalization disabled. The interval is mapped to each model's native time coordinate. Timing covers text prefill, denoising, VAE decoding where applicable, and CPU image conversion. Image-token throughput is the final spatial token count divided by end-to-end latency; repeated denoising forwards do not count as newly emitted tokens.

Tab. 9 shows that UniSpark reduces median generation latency relative to BAGEL from 11.76 to 7.27 seconds at 512^2 and from 45.80 to 14.16 seconds at 1024^2, giving $1.62\times$ and $3.23\times$ speedups. SenseNova-U1 is faster at these two resolutions, but UniSpark is faster at 2048^2 (51.48 versus 58.34 seconds). This advantage comes with higher peak allocated memory: 62.74 GB versus 37.39 GB at 2048^2. The results therefore show a resolution-dependent latency benefit with a memory cost.

Image-token throughput also depends on the representation size. At 1024^2, BAGEL uses 4,096 image tokens, compared with 1,024 for UniSpark and SenseNova-U1, so its higher tokens/s does not imply faster image generation. At 2048^2, the latter two models both use 4,096 tokens, and UniSpark reaches 79.56 versus 70.20 tokens/s. SenseNova-U1's two smaller settings are outside its recommended trained resolution buckets, so

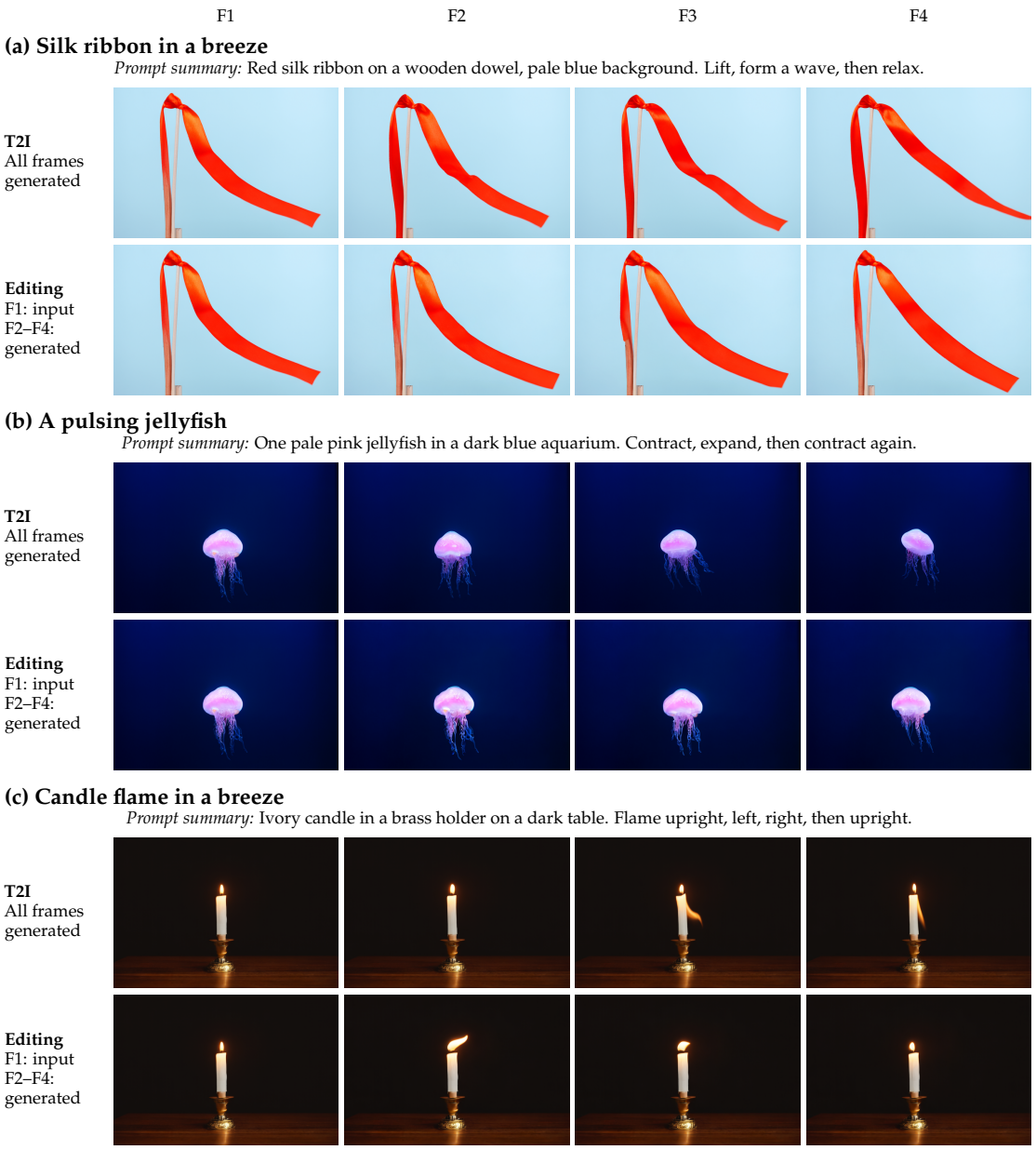


Figure 11: **Text-only generation (top) and first-frame-conditioned editing (bottom).** Generation predicts F1–F4 jointly, whereas editing conditions on F1 to predict F2–F4, illustrating visual continuity under two conditioning settings.

these measurements characterize fixed workloads rather than matched image quality.

For visual understanding, we use the same four image–question pairs with 256 or 1,024 visual tokens per image, checked at the language-model input. To match these budgets, BAGEL receives 224×224 or 448×448 images, while UniSpark and SenseNova-U1 receive 512×512 or 1024×1024 images. Each run makes exactly 128 greedy token predictions, ignoring early termination. Native tokenizers and chat templates are retained; any reasoning text produced by SenseNova-U1 counts toward this fixed budget.

Images are loaded and resized before timing. Total latency includes native preprocessing, vision encoding, multimodal prefill, and text decoding. Decoding throughput is $127/(t_{128} - t_1)$, where t_i is a CUDA event recorded after the output projection for prediction i. First-logit latency measures request start to t_1, before token selection; it is not user-visible time to first token.

Type	Model	Add	Adjust	Extract	Replace	Remove	Back.	Style	Hybrid	Action	Overall↑
Gen. Only	FLUX.1 Kontext (2025)	4.25	4.15	2.35	4.56	3.57	4.26	4.57	3.68	4.63	4.00
	Step1X-Edit (2025)	3.88	3.14	1.76	3.40	2.41	3.16	4.63	2.64	2.52	3.06
	Qwen-Image-Edit (2025)	4.32	4.36	4.04	4.64	4.52	4.37	4.84	3.39	4.71	4.35
	Z-Image-Edit (2025)	4.40	4.14	4.30	4.57	4.13	4.14	4.85	3.63	4.50	4.30
Unified	BAGEL (2025)	3.56	3.31	1.70	3.30	2.62	3.24	4.49	2.38	4.17	3.20
	OmniGen2 (2025b)	3.57	3.06	1.77	3.74	3.20	3.57	4.81	2.52	4.68	3.44
	Lumina-DiMOO (2025a)	3.41	2.38	1.90	3.26	2.21	2.11	4.19	2.26	3.17	2.77
	InternVL-U (2026)	4.13	3.40	2.27	4.13	3.39	3.84	4.77	3.03	4.05	3.67
	LLaDA2.0-Uni (2026)	3.76	4.16	2.40	4.04	3.82	4.07	4.60	3.97	4.42	3.92
	UniSpark†	3.72	2.54	1.68	3.51	3.52	3.47	4.05	1.62	3.44	3.13

Table 8: **Instruction-based image editing on ImgEdit.** Back.: background editing; †: Qwen3-VL reference scores (different judge).

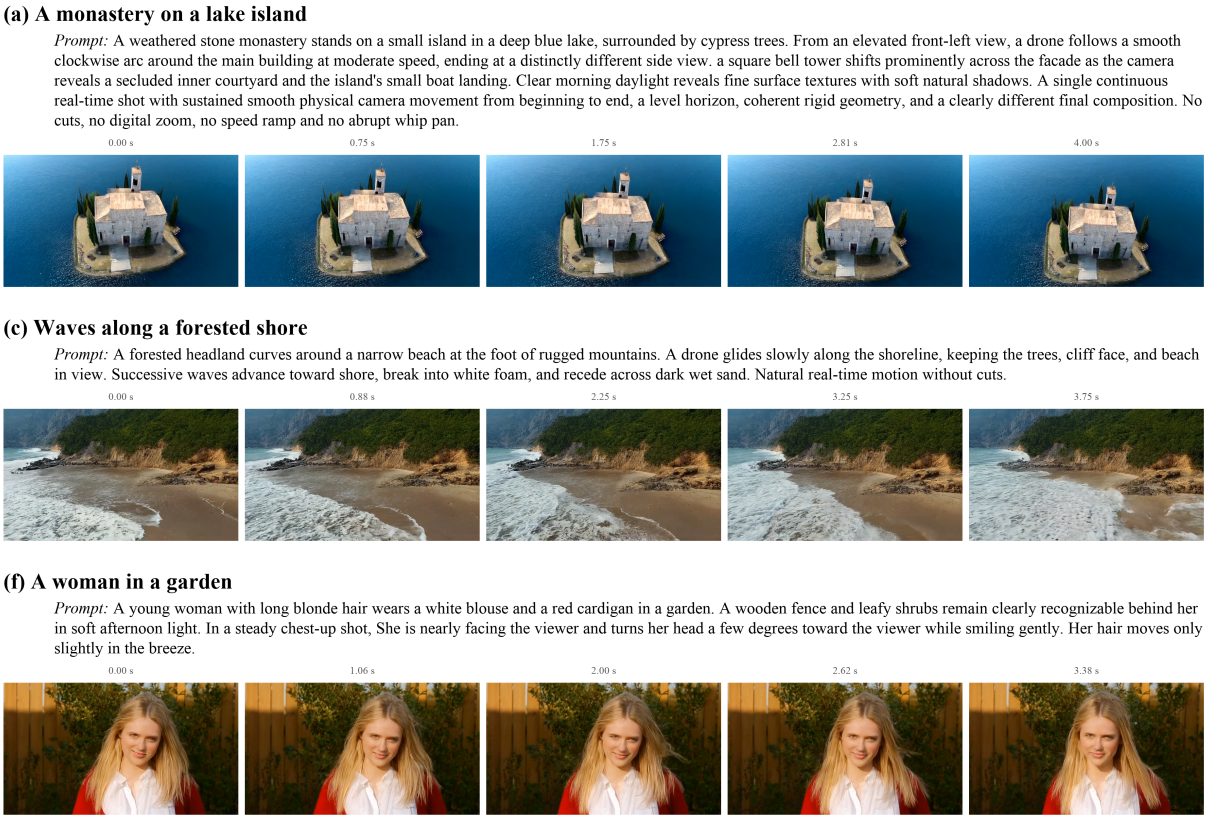


Figure 12: **Qualitative examples of video generation.** Chronological frames illustrate text-conditioned scene evolution, including camera movement, waves, and subtle human motion.

Tab. 10 places UniSpark between BAGEL and SenseNova-U1 in decoding throughput and total latency at both visual-token budgets. At 1,024 tokens, it decodes at 22.19 tokens/s and finishes in 6.02 seconds, compared with 36.60 tokens/s and 3.76 seconds for BAGEL, and 18.09 tokens/s and 7.32 seconds for SenseNova-U1. Increasing the visual budget from 256 to 1,024 tokens changes its total latency only from 5.94 to 6.02 seconds, although first-logit latency rises from 0.143 to 0.292 seconds. For these fixed 128-token outputs, decoding therefore accounts for most of the request time. Peak allocated memory is slightly higher than SenseNova-U1 and remains above BAGEL at both budgets. These measurements describe computational cost rather than answer quality.

The two tables distinguish generation speed from autoregressive decoding speed: UniSpark reduces gen-

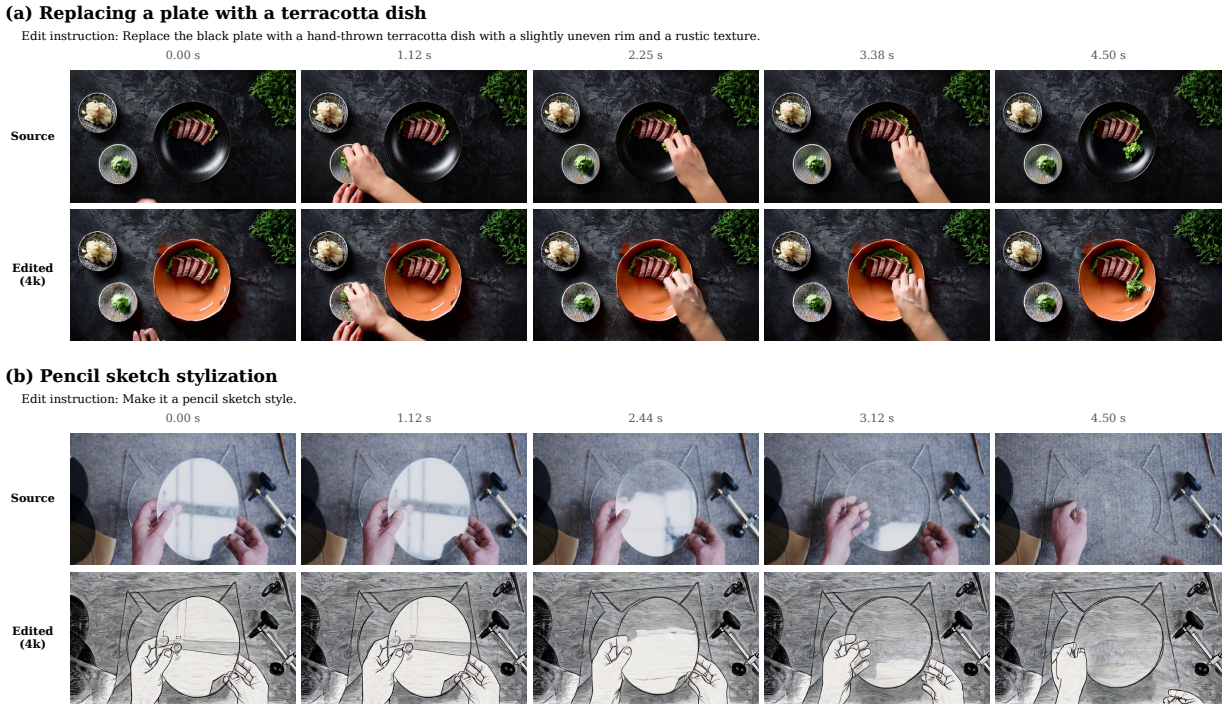


Figure 13: **Qualitative examples of video editing.** Temporally aligned source and edited frames illustrate object replacement and global stylization, allowing comparison of edit adherence and source-content preservation.

Resolution	Model	# LLM Params	Image tokens/s↑	Median latency (s)↓	P95 latency (s)↓	Peak allocated memory (GB)↓
512 × 512	BAGEL (2025)	7B MoT	87.09	11.76	11.77	30.47
	SenseNova-U1 (2026)	8B MoT	39.80	6.43	6.52	35.49
	UniSpark	9B MoT	35.19	7.27	7.36	37.39
1024 × 1024	BAGEL (2025)	7B MoT	89.44	45.80	45.81	33.29
	SenseNova-U1 (2026)	8B MoT	73.90	13.86	13.91	35.84
	UniSpark	9B MoT	72.33	14.16	14.21	42.47
2048 × 2048	BAGEL (2025)	7B MoT	–	–	–	–
	SenseNova-U1 (2026)	8B MoT	70.20	58.34	58.36	37.39
	UniSpark	9B MoT	79.56	51.48	51.51	62.74

Table 9: **Image generation efficiency across output resolutions.**

eration latency relative to BAGEL at the shared resolutions, whereas BAGEL remains faster and uses less allocated memory for understanding. Matched workloads control sequence lengths, but model sizes, native tokenization, vision encoders, and implementations still differ. These system-level measurements do not isolate the effect of hybrid attention.

Visual tokens	Model	# LLM Params	Input	Text decode tokens/s↑	First logits (s)↓	Median latency (s)↓	Peak allocated memory (GB)↓
256	BAGEL (2025)	7B MoT	224 × 224	36.09	0.176	3.70	29.58
	SenseNova-U1 (2026)	8B MoT	512 × 512	17.92	0.165	7.26	35.32
	UniSpark	9B MoT	512 × 512	21.91	0.143	5.94	35.73
1,024	BAGEL (2025)	7B MoT	448 × 448	36.60	0.263	3.76	29.77
	SenseNova-U1 (2026)	8B MoT	1024 × 1024	18.09	0.289	7.32	35.85
	UniSpark	9B MoT	1024 × 1024	22.19	0.292	6.02	36.03

Table 10: **Visual understanding efficiency at matched visual-token budgets.**

7 Conclusion

In this work, we present UniSpark, a unified multimodal model that explores hybrid attention for understanding and generation within a shared Mixture-of-Transformer-Experts architecture. By combining full attention with recurrent linear attention, UniSpark brings detailed visual interactions and compact context representations into the same backbone. Joint language modeling and visual flow matching connect image understanding, generation, and instruction-based editing, while clean visual context links successive outputs in interleaved sequences. Diverse multimodal supervision and progressive resolution training extend this framework to a broad range of tasks and image resolutions up to 2K. Experiments demonstrate strong multimodal understanding, and visual examples illustrate generation, editing, and interleaved prediction within the unified framework. Our results provide empirical evidence that hybrid attention can support both multimodal understanding and visual generation within a unified model. The context-handling design, training recipe, and evaluation results offer a practical reference for further studies of hybrid architectures for unified multimodal modeling.

References

Qingyan Bai, Qiuyu Wang, Hao Ouyang, Yue Yu, Hanlin Wang, Wen Wang, Ka Leong Cheng, Shuailei Ma, Yanhong Zeng, Zichen Liu, Yinghao Xu, Yujun Shen, and Qifeng Chen. Scaling instruction-based video editing with a high-quality synthetic dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025. doi: 10.48550/arXiv.2502.13923. URL https://arxiv.org/abs/2502.13923.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*, 2015. URL https://arxiv.org/abs/1506.03099.

Black Forest Labs. FLUX.1 [dev] Model Card. https://huggingface.co/black-forest-labs/FLUX.1-dev, 2024.

Black Forest Labs, Stephen Batifol, Andreas Blattmann, Frederic Boesel, Saksham Consul, Cyril Diagne, Tim Dockhorn, Jack English, Zion English, Patrick Esser, et al. FLUX.1 Kontext: Flow Matching for In-Context Image Generation and Editing in Latent Space. *arXiv preprint arXiv:2506.15742*, 2025. URL https://arxiv.org/abs/2506.15742.

Chameleon Team. Chameleon: Mixed-Modal Early-Fusion Foundation Models. *arXiv preprint arXiv:2405.09818*, 2024. URL https://arxiv.org/abs/2405.09818.

Junsong Chen, Jincheng Yu, Yitong Li, Shuchen Xue, Haozhe Liu, Jingyu Xin, Yuyang Zhao, Tian Ye, Zhangjie Wu, Zian Wang, Daquan Zhou, Ping Luo, Song Han, and Enze Xie. SANA-Video 2.0: Hybrid linear attention with attention residuals for efficient video generation. *arXiv preprint arXiv:2607.21553*, 2026. doi: 10.48550/arXiv.2607.21553. URL https://arxiv.org/abs/2607.21553.

Tianqi Chen, Bing Xu, Chiyuan Zhang, and Carlos Guestrin. Training deep nets with sublinear memory cost. *arXiv preprint arXiv:1604.06174*, 2016. URL https://arxiv.org/abs/1604.06174.

Xiaokang Chen, Zhiyu Wu, Xingchao Liu, Zizheng Pan, Wen Liu, Zhenda Xie, Xingkai Yu, and Chong Ruan.

Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling. *arXiv preprint arXiv:2501.17811*, 2025. URL https://arxiv.org/abs/2501.17811.

Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling. *arXiv preprint arXiv:2412.05271*, 2024. URL https://arxiv.org/abs/2412.05271.

Chaorui Deng, Deyao Zhu, Kunchang Li, Chenhui Gou, Feng Li, Zeyu Wang, Shu Zhong, Weihao Yu, Xiaonan Nie, Ziang Song, et al. Emerging properties in unified multimodal pretraining. *arXiv preprint arXiv:2505.14683*, 2025. doi: 10.48550/arXiv.2505.14683. URL https://arxiv.org/abs/2505.14683.

Haiwen Diao, Penghao Wu, Hanming Deng, Jiahao Wang, Shihao Bai, Silei Wu, Weichen Fan, Wenjie Ye, Wenwen Tong, Xiangyu Fan, Yan Li, Yubo Wang, Zhijie Cao, Zhiqian Lin, Zhitao Yang, Zhongang Cai, Yuwei Niu, Yue Zhu, Bo Liu, Chengguang Lv, Haojia Yu, Haozhe Xie, Hongli Wang, Jianan Fan, Jiaqi Li, Jiefan Lu, Jingcheng Ni, Junxiang Xu, Kaihuan Liang, Lianqiang Shi, Linjun Dai, Linyan Wang, Oscar Qian, Peng Gao, Pengfei Liu, Qingping Sun, Rui Shen, Ruisi Wang, Shengnan Ma, Shuang Yang, Siyi Xie, Siying Li, Tianbo Zhong, Xiangli Kong, Xuanke Shi, Yang Gao, Yongqiang Yao, Yves Wang, Zhengqi Bai, Zhengyu Lin, Zixin Yin, Wenxiu Sun, Ruihao Gong, Quan Wang, Lewei Lu, Lei Yang, Ziwei Liu, and Dahua Lin. SenseNova-U1: Unifying Multimodal Understanding and Generation with NEO-unify Architecture. *arXiv preprint arXiv:2605.12500*, 2026. doi: 10.48550/arXiv.2605.12500. URL https://arxiv.org/abs/2605.12500.

Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Zhenyu Qiu, Wei Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, and Rongrong Ji. MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models. *arXiv preprint arXiv:2306.13394*, 2023. doi: 10.48550/arXiv.2306.13394. URL https://arxiv.org/abs/2306.13394v1.

Yu Gao, Lixue Gong, Qiushan Guo, Xiaoxia Hou, Zhichao Lai, Fanshi Li, Liang Li, Xiaochen Lian, Chao Liao, Liyang Liu, et al. Seedream 3.0 Technical Report. *arXiv preprint arXiv:2504.11346*, 2025. URL https://arxiv.org/abs/2504.11346.

Yuying Ge, Sijie Zhao, Jinguo Zhu, Yixiao Ge, Kun Yi, Lin Song, Chen Li, Xiaohan Ding, and Ying Shan. SEED-X: Multimodal Models with Unified Multi-granularity Comprehension and Generation. *arXiv preprint arXiv:2404.14396*, 2024. URL https://arxiv.org/abs/2404.14396.

Dhruba Ghosh, Hanna Hajishirzi, and Ludwig Schmidt. GenEval: An Object-Focused Framework for Evaluating Text-to-Image Alignment. *arXiv preprint arXiv:2310.11513*, 2023. URL https://arxiv.org/abs/2310.11513.

Xiaoyang Han, Jianhua Li, Kewang Deng, Zukai Chen, Xuanke Shi, Sihan Wang, Boxuan Li, Linyan Wang, Siyi Xie, Xin You, Jinsheng Quan, Zhongang Cai, Haiwen Diao, Ziwei Liu, Lei Yang, Dahua Lin, and Quan Wang. Vision as Unified Multimodal Generation. *arXiv preprint arXiv:2607.06560*, 2026. doi: 10.48550/arXiv.2607.06560. URL https://arxiv.org/abs/2607.06560.

Ali Hatamizadeh, Yejin Choi, and Jan Kautz. Gated DeltaNet-2: Decoupling Erase and Write in Linear Attention. *arXiv preprint arXiv:2605.22791*, 2026. doi: 10.48550/arXiv.2605.22791. URL https://arxiv.org/abs/2605.22791.

Xiwei Hu, Rui Wang, Yixiao Fang, Bin Fu, Pei Cheng, and Gang Yu. ELLA: Equip Diffusion Models with LLM for Enhanced Semantic Alignment. *arXiv preprint arXiv:2403.05135*, 2024. URL https://arxiv.org/abs/2403.05135.

Inclusion AI, Tiwei Bie, Haoxing Chen, Tieyuan Chen, Zhenglin Cheng, Long Cui, Kai Gan, Zhicheng Huang, Zhenzhong Lan, Haoquan Li, et al. LLaDA2.0-Uni: Unifying Multimodal Understanding and Generation with Diffusion Large Language Model. *arXiv preprint arXiv:2604.20796*, 2026. URL https://arxiv.org/abs/2604.20796.

Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pp. 5156–5165, 2020. URL https://proceedings.mlr.press/v119/katharopoulos20a.html.

Kimi Team, Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, Cheng Chen, Chenlin Zhang,

Chenzhuang Du, Chu Wei, et al. Kimi-VL Technical Report. *arXiv preprint arXiv:2504.07491*, 2025a. URL https://arxiv.org/abs/2504.07491.

Kimi Team, Yu Zhang, Zongyu Lin, Xingcheng Yao, Jiaxi Hu, Fanqing Meng, Chengyin Liu, Xin Men, Songlin Yang, Zhiyuan Li, Wentao Li, Enzhe Lu, Weizhou Liu, Yanru Chen, Weixin Xu, Longhui Yu, Yejie Wang, Yu Fan, Longguang Zhong, Enming Yuan, Dehao Zhang, Yizhi Zhang, T. Y. Liu, Haiming Wang, Shengjun Fang, Weiran He, Shaowei Liu, Yiwei Li, Jianlin Su, Jiezhong Qiu, Bo Pang, Junjie Yan, Zhejun Jiang, Weixiao Huang, Bohong Yin, Jiacheng You, Chu Wei, Zhengtao Wang, Chao Hong, Yutian Chen, Guanduo Chen, Yucheng Wang, Huabin Zheng, Feng Wang, Yibo Liu, Mengnan Dong, Zheng Zhang, Siyuan Pan, Wenhao Wu, Yuhao Wu, Longyu Guan, Jiawen Tao, Guohong Fu, Xinran Xu, Yuzhi Wang, Guokun Lai, Yuxin Wu, Xinyu Zhou, Zhilin Yang, and Yulun Du. Kimi Linear: An Expressive, Efficient Attention Architecture. *arXiv preprint arXiv:2510.26692*, 2025b. doi: 10.48550/arXiv.2510.26692. URL https://arxiv.org/abs/2510.26692.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. LLaVA-OneVision: Easy Visual Task Transfer. *arXiv preprint arXiv:2408.03326*, 2024. URL https://arxiv.org/abs/2408.03326.

Ruibang Li, Guan Luo, Yiwei Zhang, Jin Gao, Bing Li, and Weiming Hu. SoLA-Vision: Fine-grained layer-wise linear softmax hybrid attention. *arXiv preprint arXiv:2601.11164*, 2026. doi: 10.48550/arXiv.2601.11164. URL https://arxiv.org/abs/2601.11164.

Weixin Liang, Lili Yu, Liang Luo, Srinivasan Iyer, Ning Dong, Chunting Zhou, Gargi Ghosh, Mike Lewis, Wen-tau Yih, Luke Zettlemoyer, and Xi Victoria Lin. Mixture-of-transformers: A sparse and scalable architecture for multi-modal foundation models. *Transactions on Machine Learning Research*, 2025. doi: 10.48550/arXiv.2411.04996. URL https://arxiv.org/abs/2411.04996.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.02747.

Shiyu Liu, Yucheng Han, Peng Xing, Fukun Yin, Rui Wang, Wei Cheng, Jiaqi Liao, Yingming Wang, Honghao Fu, Chunrui Han, et al. Step1X-Edit: A Practical Framework for General Image Editing. *arXiv preprint arXiv:2504.17761*, 2025. URL https://arxiv.org/abs/2504.17761.

Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. MMBench: Is Your Multi-Modal Model an All-around Player? In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024. URL https://arxiv.org/abs/2307.06281.

Zhiheng Liu, Weiming Ren, Xiaoke Huang, Shoufa Chen, Tianhong Li, Mengzhao Chen, Yatai Ji, Sen He, Jonas Schult, Belinda Zeng, Tao Xiang, Wenhu Chen, Ping Luo, Luke Zettlemoyer, and Yuren Cong. Tuna-2: Pixel Embeddings Beat Vision Encoders for Multimodal Understanding and Generation. *arXiv preprint arXiv:2604.24763*, 2026. doi: 10.48550/arXiv.2604.24763. URL https://arxiv.org/abs/2604.24763.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://arxiv.org/abs/1711.05101.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts. *arXiv preprint arXiv:2310.02255*, 2023. URL https://arxiv.org/abs/2310.02255.

Meituan LongCat Team, Hanghang Ma, Haoxian Tan, Jiale Huang, Junqiang Wu, Jun-Yan He, Lishuai Gao, Songlin Xiao, Xiaoming Wei, Xiaoqi Ma, et al. LongCat-Image Technical Report. *arXiv preprint arXiv:2512.07584*, 2025. URL https://arxiv.org/abs/2512.07584.

Boyuan Meng, Peihua Bao, Hong Liu, Xiaowei Zhu, Chao Wang, Gen Li, and Zhenxuan Pan. HARTS: Efficient Agentic Reinforcement Learning for Hybrid-Attention Models over Arbitrary Rollout Trees. *arXiv preprint arXiv:2608.28158*, 2026. doi: 10.48550/arXiv.2608.28158. URL https://arxiv.org/abs/2608.28158.

William Merrill, Yanhong Li, Tyler Romero, Anej Svete, Caia Costello, Pradeep Dasigi, Dirk Groeneveld, David Heineman, Bailey Kuehl, Nathan Lambert, Chuan Li, Kyle Lo, Saumya Malik, DJ Matusz, Benjamin Minixhofer, Jacob Morrison, Luca Soldaini, Finbarr Timbers, Pete Walsh, Noah A. Smith, Hannaneh Hajishirzi, and Ashish Sabharwal. Olmo Hybrid: From Theory to Practice and Back. *arXiv preprint arXiv:2604.03444*, 2026. doi: 10.48550/arXiv.2604.03444. URL https://arxiv.org/abs/2604.03444.

OpenGVLab. InternVL2 Model Card. https://huggingface.co/OpenGVLab/InternVL2-8B, 2024.

Xichen Pan, Satya Narayan Shukla, Aashu Singh, Zhuokai Zhao, Shlok Kumar Mishra, Jialiang Wang, Zhiyang Xu, Jiuhai Chen, Kunpeng Li, Felix Juefei-Xu, et al. Transfer between Modalities with MetaQueries. *arXiv preprint arXiv:2504.06256*, 2025. URL https://arxiv.org/abs/2504.06256.

Wujian Peng, Lingchen Meng, Yuxuan Cai, Xianwei Zhuang, Yuhuan Yang, Rongyao Fang, Chenfei Wu, Junyang Lin, Zuxuan Wu, and Shuai Bai. Unified Multimodal Autoregressive Modeling with Shared Context-Visual Tokenizer is Key to Unification. *arXiv preprint arXiv:2606.18249*, 2026. doi: 10.48550/arXiv.2606.18249. URL https://arxiv.org/abs/2606.18249.

Liao Qu, Huichao Zhang, Yiheng Liu, Xu Wang, Yi Jiang, Yiming Gao, Hu Ye, Daniel K. Du, Zehuan Yuan, and Xinglong Wu. TokenFlow: Unified Image Tokenizer for Multimodal Understanding and Generation. *arXiv preprint arXiv:2412.03069*, 2024. URL https://arxiv.org/abs/2412.03069.

Qwen Team. Qwen-Image-Edit Model Card. https://huggingface.co/Qwen/Qwen-Image-Edit, 2025.

Qwen Team. Qwen3.5-9B model card. Hugging Face, 2026. URL https://huggingface.co/Qwen/Qwen3.5-9B. Accessed September 6, 2026.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2112.10752.

Weijia Shi, Xiaochuang Han, Chunting Zhou, Weixin Liang, Xi Victoria Lin, Luke Zettlemoyer, and Lili Yu. LMFusion: Adapting Pretrained Language Models for Multimodal Generation. *arXiv preprint arXiv:2412.15188*, 2024. URL https://arxiv.org/abs/2412.15188.

Zhentao Tan, Wei Chen, Jingyi Shen, Yao Liu, Xu Shen, Yue Wu, and Jieping Ye. HydraHead: From Head-Level Functional Heterogeneity to Specialized Attention Hybridization. *arXiv preprint arXiv:2606.20097*, 2026. doi: 10.48550/arXiv.2606.20097. URL https://arxiv.org/abs/2606.20097.

Tencent Hunyuan Foundation Model Team. HunyuanImage 3.0 Technical Report. *arXiv preprint arXiv:2509.23951*, 2025. URL https://arxiv.org/abs/2509.23951.

Changyao Tian, Danni Yang, Guanzhou Chen, Erfei Cui, Zhaokai Wang, Yuchen Duan, Penghao Yin, Sitao Chen, Ganlin Yang, Mingxin Liu, et al. InternVL-U: Democratizing Unified Multimodal Models for Understanding, Reasoning, Generation and Editing. *arXiv preprint arXiv:2603.09877*, 2026. URL https://arxiv.org/abs/2603.09877.

Shengbang Tong, David Fan, Jiachen Zhu, Yunyang Xiong, Xinlei Chen, Koustuv Sinha, Michael Rabbat, Yann LeCun, Saining Xie, and Zhuang Liu. MetaMorph: Multimodal Understanding and Generation via Instruction Tuning. *arXiv preprint arXiv:2412.14164*, 2024a. URL https://arxiv.org/abs/2412.14164.

Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. *arXiv preprint arXiv:2401.06209*, 2024b. doi: 10.48550/arXiv.2401.06209. URL https://arxiv.org/abs/2401.06209.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017. URL https://arxiv.org/abs/1706.03762.

Wan Team. Wan2.2: Open and Advanced Large-Scale Video Generative Models. https://github.com/Wan-Video/Wan2.2, 2025.

Chunwei Wang, Guansong Lu, Junwei Yang, Runhui Huang, Jianhua Han, Lu Hou, Wei Zhang, and Hang Xu. ILLUME: Illuminating Your LLMs to See, Draw, and Self-Enhance. *arXiv preprint arXiv:2412.06673*, 2024a. URL https://arxiv.org/abs/2412.06673.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. *arXiv preprint arXiv:2409.12191*, 2024b. URL https://arxiv.org/abs/2409.12191.

Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yueze Wang, Zhen Li, Qiying Yu, et al. Emu3: Next-Token Prediction Is All You Need. *arXiv preprint arXiv:2409.18869*, 2024c. URL https://arxiv.org/abs/2409.18869.

Ronald J. Williams and David Zipser. A learning algorithm for continually running fully recurrent neural networks. *Neural Computation*, 1(2):270–280, 1989. doi: 10.1162/neco.1989.1.2.270.

Chenfei Wu, Jiahao Li, Jingren Zhou, Junyang Lin, Kaiyuan Gao, Kun Yan, Sheng-ming Yin, Shuai Bai, Xiao Xu, Yilei Chen, et al. Qwen-Image Technical Report. *arXiv preprint arXiv:2508.02324*, 2025a. URL https://arxiv.org/abs/2508.02324.

Chengyue Wu, Xiaokang Chen, Zhiyu Wu, Yiyang Ma, Xingchao Liu, Zizheng Pan, Wen Liu, Zhenda Xie, Xingkai Yu, Chong Ruan, et al. Janus: Decoupling Visual Encoding for Unified Multimodal Understanding and Generation. *arXiv preprint arXiv:2410.13848*, 2024a. URL https://arxiv.org/abs/2410.13848.

Chenyuan Wu, Pengfei Zheng, Ruiran Yan, Shitao Xiao, Xin Luo, Yueze Wang, Wanli Li, Xiyan Jiang, Yexin Liu, Junjie Zhou, et al. OmniGen2: Towards Instruction-Aligned Multimodal Generation. *arXiv preprint arXiv:2506.18871*, 2025b. URL https://arxiv.org/abs/2506.18871.

Penghao Wu, Haiwen Diao, Weichen Fan, Lewei Lu, Dahua Lin, and Ziwei Liu. Uncovering Understanding-Generation Synergy in Native Unified Multimodal Models: From Representation, Task to System. *arXiv preprint arXiv:2609.01607*, 2026. doi: 10.48550/arXiv.2609.01607. URL https://arxiv.org/abs/2609.01607.

Yecheng Wu, Zhuoyang Zhang, Junyu Chen, Haotian Tang, Dacheng Li, Yunhao Fang, Ligeng Zhu, Enze Xie, Hongxu Yin, Li Yi, et al. VILA-U: a Unified Foundation Model Integrating Visual Understanding and Generation. *arXiv preprint arXiv:2409.04429*, 2024b. URL https://arxiv.org/abs/2409.04429.

Zhiyu Wu, Xiaokang Chen, Zizheng Pan, Xingchao Liu, Wen Liu, Damai Dai, Huazuo Gao, Yiyang Ma, Chengyue Wu, Bingxuan Wang, et al. DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. *arXiv preprint arXiv:2412.10302*, 2024c. URL https://arxiv.org/abs/2412.10302.

Jinheng Xie, Weijia Mao, Zechen Bai, David Junhao Zhang, Weihao Wang, Kevin Qinghong Lin, Yuchao Gu, Zhijie Chen, Zhenheng Yang, and Mike Zheng Shou. Show-o: One single transformer to unify multimodal understanding and generation. *arXiv preprint arXiv:2408.12528*, 2024a. doi: 10.48550/arXiv.2408.12528. URL https://arxiv.org/abs/2408.12528.

Jinheng Xie, Zhenheng Yang, and Mike Zheng Shou. Show-o2: Improved native unified multimodal models. *arXiv preprint arXiv:2506.15564*, 2025. doi: 10.48550/arXiv.2506.15564. URL https://arxiv.org/abs/2506.15564.

Rongchang Xie, Chen Du, Ping Song, and Chang Liu. MUSE-VL: Modeling Unified VLM through Semantic Discrete Encoding. *arXiv preprint arXiv:2411.17762*, 2024b. URL https://arxiv.org/abs/2411.17762.

Yi Xin, Qi Qin, Siqi Luo, Kaiwen Zhu, Juncheng Yan, Yan Tai, Jiayi Lei, Yuewen Cao, Keqi Wang, Yibin Wang, et al. Lumina-DiMOO: An Omni Diffusion Large Language Model for Multi-Modal Generation and Understanding. *arXiv preprint arXiv:2510.06308*, 2025a. URL https://arxiv.org/abs/2510.06308.

Yi Xin, Juncheng Yan, Qi Qin, Zhen Li, Dongyang Liu, Shicheng Li, Victor Shea-Jay Huang, Yupeng Zhou, Renrui Zhang, Le Zhuo, et al. Lumina-mGPT 2.0: Stand-Alone AutoRegressive Image Modeling. *arXiv preprint arXiv:2507.17801*, 2025b. URL https://arxiv.org/abs/2507.17801.

Le Xue, Manli Shu, Anas Awadalla, Jun Wang, An Yan, Senthil Purushwalkam, Honglu Zhou, Viraj Prabhu, Yutong Dai, Michael S Ryoo, et al. xGen-MM (BLIP-3): A Family of Open Large Multimodal Models. *arXiv preprint arXiv:2408.08872*, 2024. URL https://arxiv.org/abs/2408.08872.

Ceyuan Yang, Zhijie Lin, Yang Zhao, Fei Xiao, Hao He, Qi Zhao, Chaorui Deng, Kunchang Li, Zihan Ding, Yuwei Guo, Fuyun Wang, Fangqi Zhu, Xiaonan Nie, Shenhan Zhu, Shanchuan Lin, Hongsheng Li, Weilin Huang, Guang Shi, and Haoqi Fan. Context Unrolling in Omni Models. *arXiv preprint arXiv:2604.21921*, 2026. doi: 10.48550/arXiv.2604.21921. URL https://arxiv.org/abs/2604.21921.

Ling Yang, Ye Tian, Bowen Li, Xinchen Zhang, Ke Shen, Yunhai Tong, and Mengdi Wang. MMaDA: Multimodal Large Diffusion Language Models. *arXiv preprint arXiv:2505.15809*, 2025a. URL https://arxiv.org/abs/2505.15809.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. In *International Conference on Learning Representations*, 2025b. doi: 10.48550/arXiv.2412.06464. URL https://arxiv.org/abs/2412.06464.

Yang Ye, Xianyi He, Zongjian Li, Bin Lin, Shenghai Yuan, Zhiyuan Yan, Bohan Hou, and Li Yuan. ImgEdit: A Unified Image Editing Dataset and Benchmark. *arXiv preprint arXiv:2505.20275*, 2025. doi: 10.48550/arXiv.2505.20275. URL https://arxiv.org/abs/2505.20275.

Zebin You, Xiaolu Zhang, Jun Zhou, Chongxuan Li, and Ji-Rong Wen. LLaDA-o: An Effective and Length-Adaptive Omni Diffusion Model. *arXiv preprint arXiv:2603.01068*, 2026. URL https://arxiv.org/abs/2603.01068.

Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. MM-Vet: Evaluating Large Multimodal Models for Integrated Capabilities. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 57730–57754. PMLR, 2024. URL https://proceedings.mlr.press/v235/yu24o.html.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, et al. MMMU: A Massive Multi-Discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. URL https://openaccess.thecvf.com/content/CVPR2024/html/Yue_MMMU_A_Massive_Multi-discipline_Multimodal_Understanding_and_Reasoning_Benchmark_for_CVPR_2024_paper.html.

Z-Image Team, Huanqia Cai, Sihan Cao, Ruoyi Du, Peng Gao, Steven Hoi, Zhaohui Hou, Shijie Huang, Dengyang Jiang, Xin Jin, Liangchen Li, et al. Z-Image: An Efficient Image Generation Foundation Model with Single-Stream Diffusion Transformer. *arXiv preprint arXiv:2511.22699*, 2025. URL https://arxiv.org/abs/2511.22699.

Huichao Zhang, Liao Qu, Yiheng Liu, Hang Chen, Yangyang Song, Yongsheng Dong, Shikun Sun, Xian Li, Xu Wang, Yi Jiang, et al. NextFlow: Unified Sequential Modeling Activates Multimodal Understanding and Generation. *arXiv preprint arXiv:2601.02204*, 2026. URL https://arxiv.org/abs/2601.02204.

Yanli Zhao, Andrew Gu, Rohan Varma, Liang Luo, Chien-Chin Huang, Min Xu, Less Wright, Hamid Shojanazeri, Myle Ott, Sam Shleifer, et al. PyTorch FSDP: Experiences on Scaling Fully Sharded Data Parallel. *arXiv preprint arXiv:2304.11277*, 2023. URL https://arxiv.org/abs/2304.11277.

Chunting Zhou, Lili Yu, Arun Babu, Kushal Tirumala, Michihiro Yasunaga, Leonid Shamis, Jacob Kahn, Xuezhe Ma, Luke Zettlemoyer, and Omer Levy. Transfusion: Predict the next token and diffuse images with one multi-modal model. *arXiv preprint arXiv:2408.11039*, 2024. doi: 10.48550/arXiv.2408.11039. URL https://arxiv.org/abs/2408.11039.

Yubo Zhu, Zhehan Kan, Jingyi Yang, Miaolin Chen, Jinbo Xing, Kai Zhu, Zijian Wang, Sheng Zhong, and Wei Tong. When Does Visual Generation Help Visual Understanding in Unified Multimodal Models? *arXiv preprint arXiv:2608.22174*, 2026. doi: 10.48550/arXiv.2608.22174. URL https://arxiv.org/abs/2608.22174.